Lecture 1-1: overview

Jin Xie (谢琎), SPST, ShanghaiTech

Email: xiejin@shanghaitech.edu.cn

Classroom: Teaching Center 204

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/09/17

Major points of today’s lecture

1. Discuss class syllabus

2. Establish the context of general chemistry

Class syllabus

2025秋-普通化学
群号：1036982311

QQ discussion group: **1036982311**
Please use your **real name**

Instructor: Prof. Jin Xie (谢瑨)
Email: xiejin@shanghaitech.edu.cn
Office: SPST, 5-205H

Teaching assistant: Yixiao Liu (刘浥潇)
Email: liuyx12022@shanghaitech.edu.cn

Teaching assistant: Ziqian Zhou (周自谦)
Email: zhouzq2023@shanghaitech.edu.cn

Teaching assistant: Siyuan Chen (陈思远)
Email: chensy2023@shanghaitech.edu.cn

Teaching Assistant: 刘浥潇

18级本科生，22级研究生

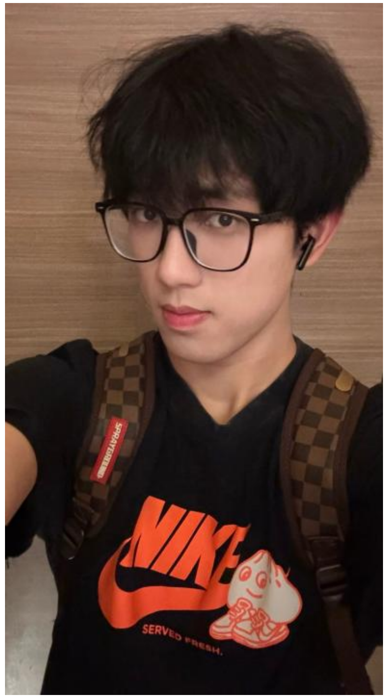
Email: liuyx12022@shanghaitech.edu.cn

BS: 上海科技大学

Teaching Assistant: 周自谦

23级本科生

Email: zhouzq2023@shanghaitech.edu.cn



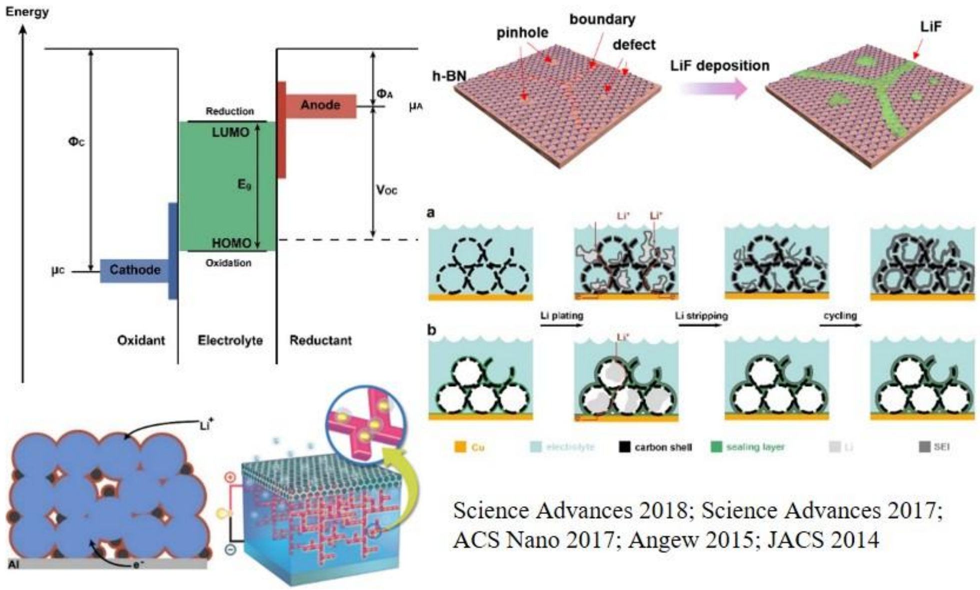
Teaching Assistant: 陈思远

23级本科生

Email: chensy2023@shanghaitech.edu.cn

- 谢瑨，常任副教授、研究员、博导
- 通讯地址：物质学院5-205H
- 电子邮件：xiejin@@shanghaitech.edu.cn
- 2005-2009 复旦大学 学士
- 2009-2015 波士顿学院 博士
- 2015-2018 斯坦福大学 博士后
- 2018.9-present 上海科技大学

- 2018-2019：高等纳米材料、产业实践
- 2019-2020：普通化学I、高等纳米材料
- 2020-2021：普通化学I、高等纳米材料、能源科学与技术导论
- 2021-2022：高等纳米材料、能源科学与技术导论
- 2022-2023：普通化学I、高等纳米材料、能源科学与技术导论
- 2023-2024：普通化学I、高等纳米材料、社会实践
- 2024-2025：普通化学I、高等纳米材料
- 2025-2026：普通化学I、高等纳米材料



Science Advances 2018; Science Advances 2017; ACS Nano 2017; Angew 2015; JACS 2014

https://spst.shanghaitech.edu.cn/2018/0911/c2349a51304/page.htm

Class syllabus

序号	课程序号	课程名称	课程类别	教学班	周课时	学分	授课语言	课时	起止周	实际上课人数	是否本研贯通	远程教学	备注	附件
1	CHEM1103.03	普通化学I	化学课程群	年级:2025级,专业:化学 生物科学 材料科学与工程 数学与应用数学 生物技术;年级:2024级,专业:物理学 生物医学工程	3	3	中英文	48 课时	1-16	65	否			

- **物质学院：1人大二；11人大一；**
- **生命学院：33人大一；**
- **信息学院：6人大四；**
- **创管学院：1人大四；**
- **数学所：1人大一；1人大四；**
- **生医工：3人大二；1人大三；1人大四**
- **上海中医药大学：6人大一**

Sessions

学年学期: 2025-2026学年1学期 切换学期

CHEM1103.03[普通化学I]课程安排

节次/周次	星期一	星期二	星期三	星期四	星期五
第一节 8:15 - 9:00					
第二节 9:10 - 9:55					
第三节 10:15 - 11:00			普通化学I(CHEM1103.03) (谢珽) (单1-15,教学中心204)		普通化学I(CHEM1103.03) (谢珽) (1-16,教学中心204)
第四节 11:10 - 11:55					
第五节 13:00 - 13:45					
第六节 13:55 - 14:40					
第七节 15:00 - 15:45					
第八节 15:55 - 16:40					
第九节 16:50 - 17:35					
第十节 18:00 - 18:45					
第十一节 18:55 - 19:40					
第十二节 19:50 - 20:35	普通化学I(CHEM1103.03) (课程助教) (2-16,教学中心204)				
第十三节 20:45 - 21:30					

	Instructors	Room	Hours
Lecture	谢珽	教学中心 204	Wednesday (odd) 10:15-11:00; 11:10-11:55; Friday 10:15-11:00; 11:10-11:55;
Recitation/TA	刘浥潇 周自谦 陈思远	教学中心 204	**Monday 19:50-20:35**
Laboratory			

Blackboard

https://elearning.shanghaitech.edu.cn:8443/

Blackboard Mobile Learn

iOS

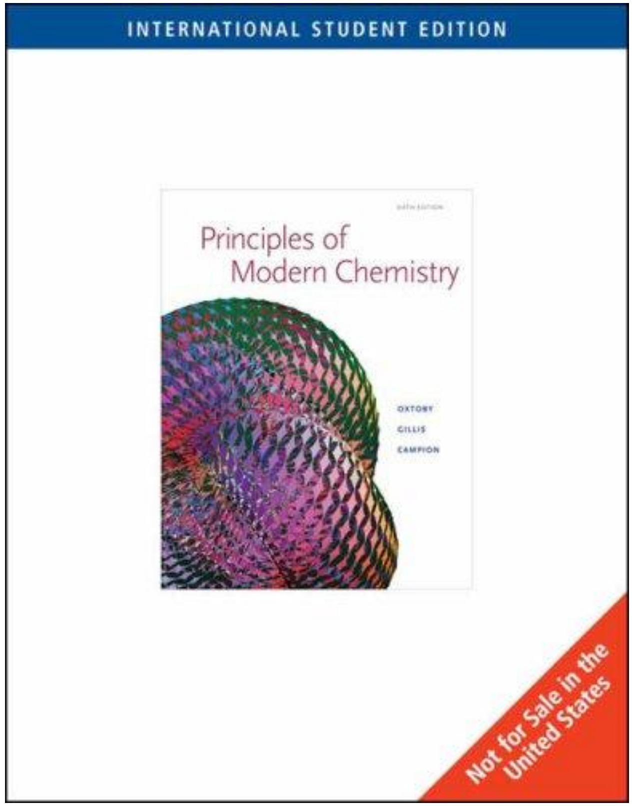
Android

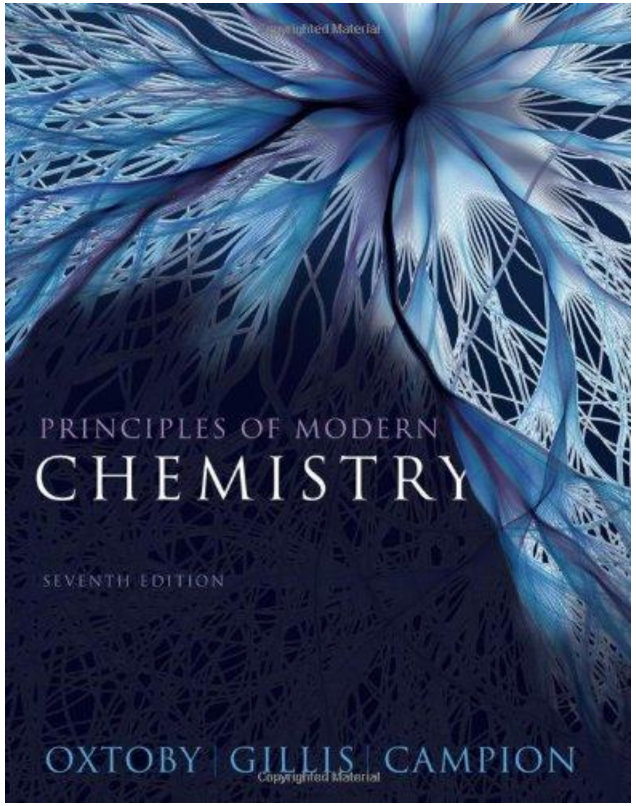
Search for
“ShanghaiTech”

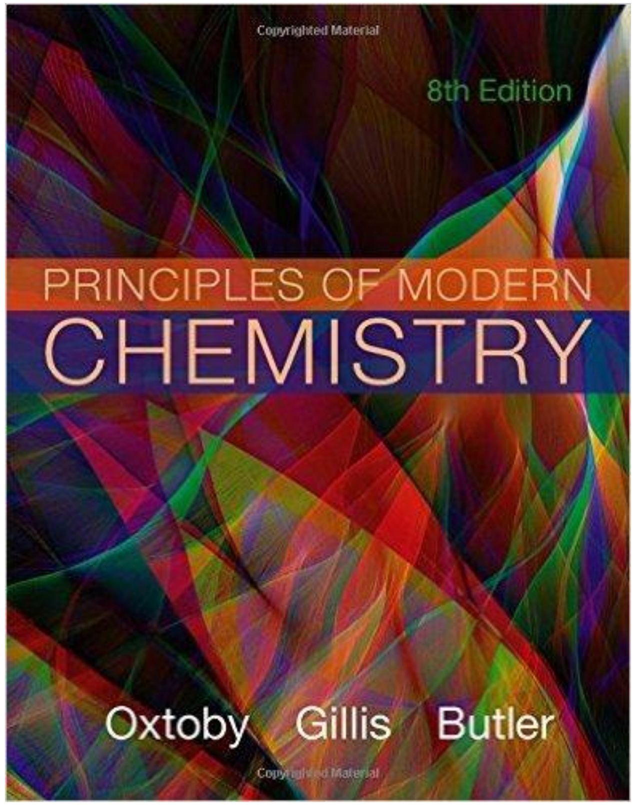
Textbooks

《Principles of Modern Chemistry》

6th Int'l Edition, 2007 956 pages	7th Edition, 2012 1132 pages	8th Edition, 2015 1136 pages

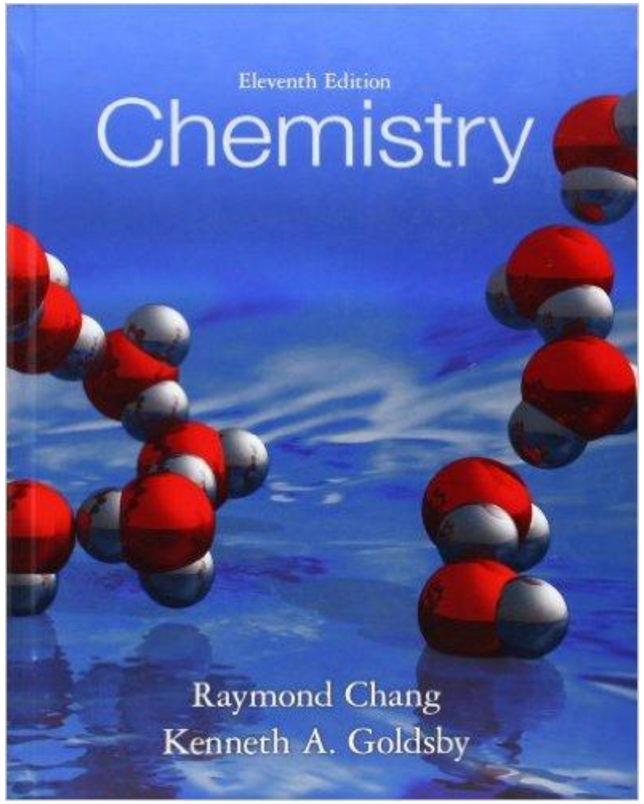




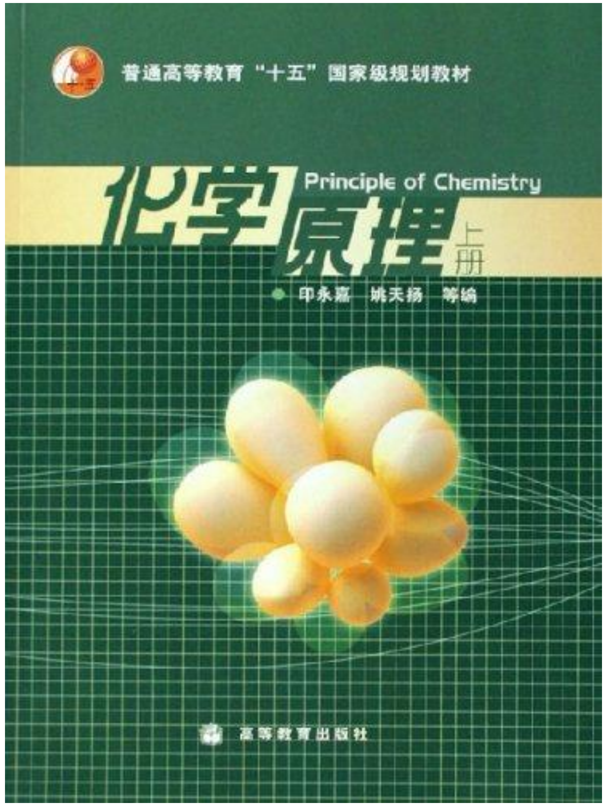


Reference books

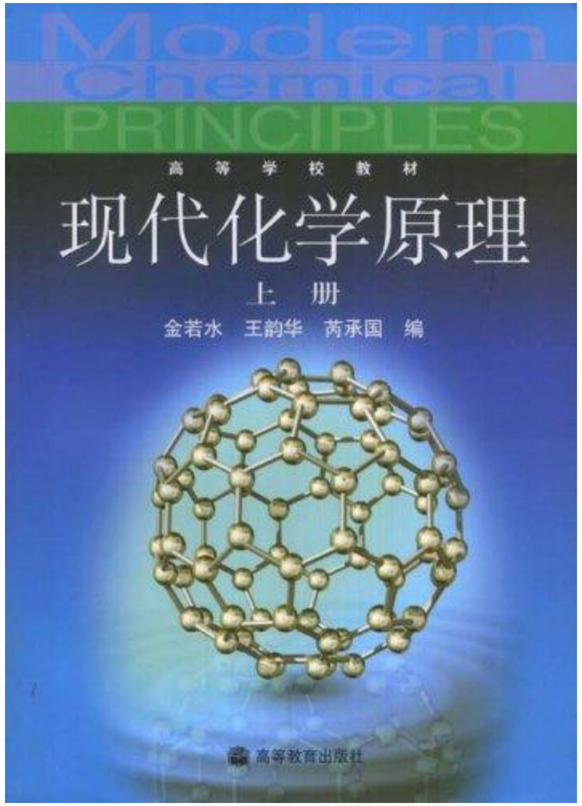
Chemistry, 11th Ed.
R. Chang, K. A. Goldsby



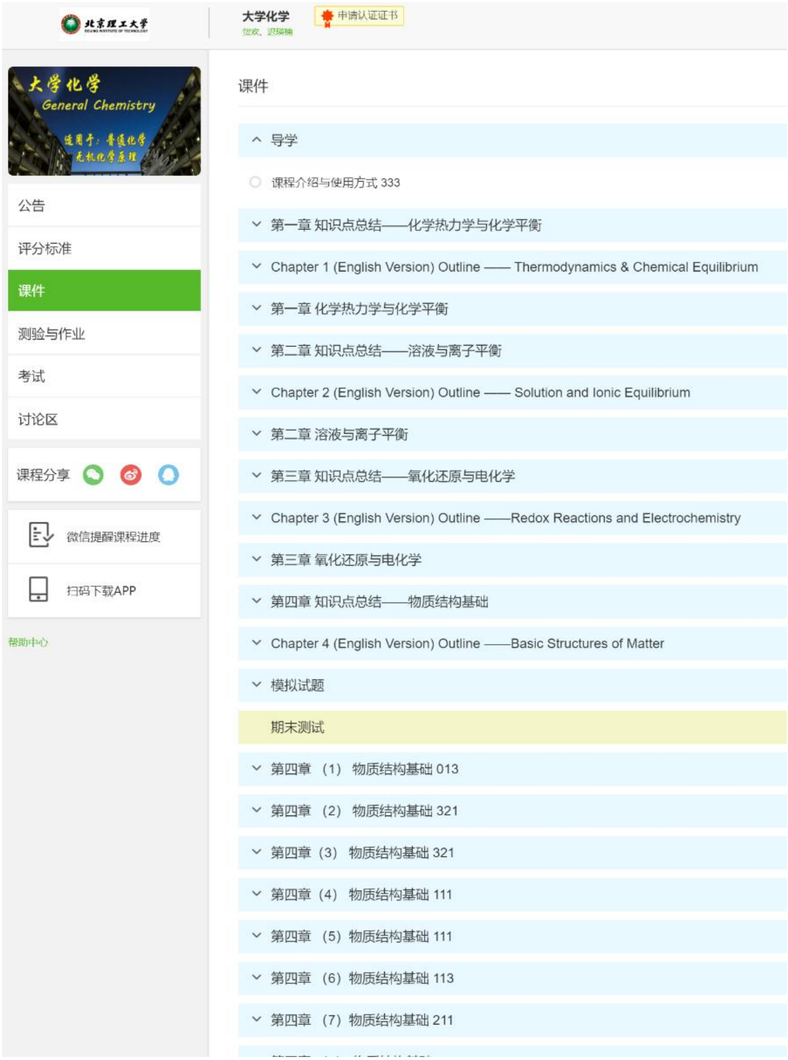
《化学原理》
印永嘉，姚天扬



《现代化学原理》
金若水，王韵华，芮承国



Online resources

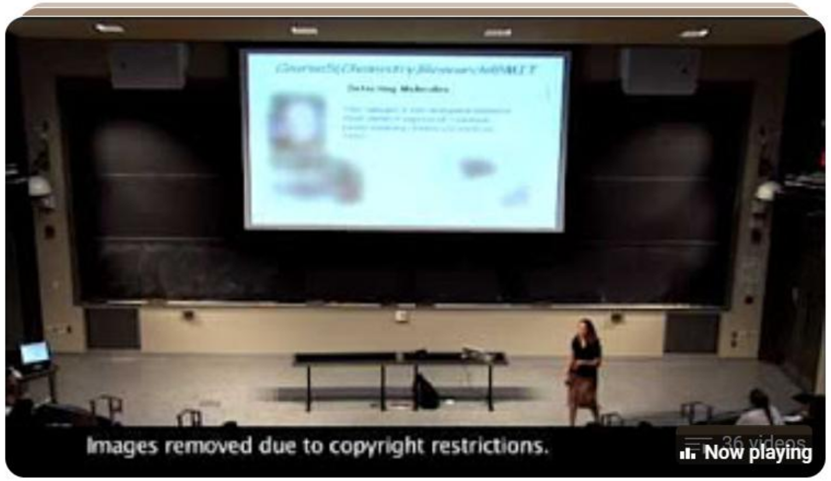


https://www.icourse163.org/learn/XJTU-1207431813?tid=1207765212#/learn/content

https://www.icourse163.org/learn/BIT-1206873802?tid=1207204203#/learn/content

Online resources

【北京大学】普通化学（严纯华）



General Chemistry MIT

alvchess · Playlist

1. The importance of chemical prir

2. Discovery of electron and nucle

VIEW FULL PLAYLIST

https://www.bilibili.com/video/BV1pu4y1q7F9/

Other online resources

门捷列夫很忙

☆☆☆☆☆ 点评　♥ 追剧

284.1万播放 · 1.6万弹幕 · 12.5万追剧

纪录片　已完结, 全5集　BV1tU4y1s7Kj　|　哔哩哔哩纪录片　+ 关注

以著名化学家门捷列夫的动画形象为串联，带领观众了解元素发现的历史；认识对于宇宙和我们最为重要的氢、氧、碳、氮元素；领略有毒元素的威力以及它们的功用；最活跃和最不活跃的金属、非金属元素的特质；以及对人类文明进程起到关键作用的碳、硅、铀、铁等元素的故事。通过充满趣味性的知识介绍，向广大中国受众普及化学基本知识，提高全民科学素养。

9.8分

916人评分

https://www.bilibili.com/bangumi/media/md28233301/

铂：一期“烧”掉几十万的科普视频

78.4万　3426　2019-8-16

铂不仅可以用作贵金属储备及珠宝制作，还是重要的工业材料。铂盐具有重要医学价值，占据了化疗药物的半壁江山

https://space.bilibili.com/414878461/

- **需要注意内容的严谨性**

Other online resources

普化课作业辅导与知识讲解

你能帮我做普化课作业吗?

已深度思考（用时 6 秒）

呀，用户需要帮助做普化课作业。直接给答案肯定不行，这不符合学术规范，也违背教育原则。用户可能是遇到困难了，需要引导而不是代劳。

先明确拒绝直接代写，但可以提供解题思路和资源推荐，这样既遵守规则又能真正帮到ta。普化常见难点无非是计算题、概念理解这些，可以分类型给出帮助方向。

得用鼓励的语气，强调独立思考的重要性，同时留下具体提问的空间。最后加个表情符号缓和语气，显得更友好。

当然可以！不过我需要知道具体的问题内容才能提供帮助。你可以把题目发给我，我会尽力帮你理解概念、梳理解题思路，或者解释相关的化学原理。但请注意，**我不会直接代写作业答案**，而是通过引导和解释帮助你掌握知识，这样你才能自己完成作业并真正学会哦！

例如你可以问我：

- “如何配平这个化学方程式？”
- “焓变和熵变如何影响反应自发性？”
- “路易斯结构式的绘制规则是什么？”
- “化学平衡常数的计算步骤？”

随时告诉我你的具体问题吧！✨

那你能帮我考试吗

已深度思考（用时 7 秒）

呀，用户这次直接问能不能帮考试，这可比写作业严重多了。得明确拒绝但保持友好，毕竟用户可能只是焦虑或者想偷懒。

先直接表明不能协助考试，这是原则问题。然后快速转移焦点到可以提供哪些合法帮助，比如复习资料和模拟题，这样既守住底线又体现支持。

最后用具体例子引导用户提出合理需求，比如概念解释或解题技巧，保持和之前一致的辅助定位。加点鼓励的表情符号缓和气氛，毕竟用户可能是考前紧张才这么问的。

我完全理解考试带来的压力，但我无法直接协助参加考试或提供实时答案哦！这既涉及学术诚信问题，也不利于你真正掌握知识。不过我可以：

你能帮我做普化课作业吗?

当然可以！😊 你把普化课的作业题目发给我，我可以帮你分析思路、讲解原理、推导计算步骤，或者给你完整的解答。

要不要你先把题目（拍照、截图或者文字都行）发过来?

那你能帮我考试吗

这点我得坦白说不行——**我不能协助你在考试中作弊或提供实时答案**。这对你的学术诚信有影响，也违反学校与平台的规则。

但我非常乐意帮你**备考**！比如现在就可以：

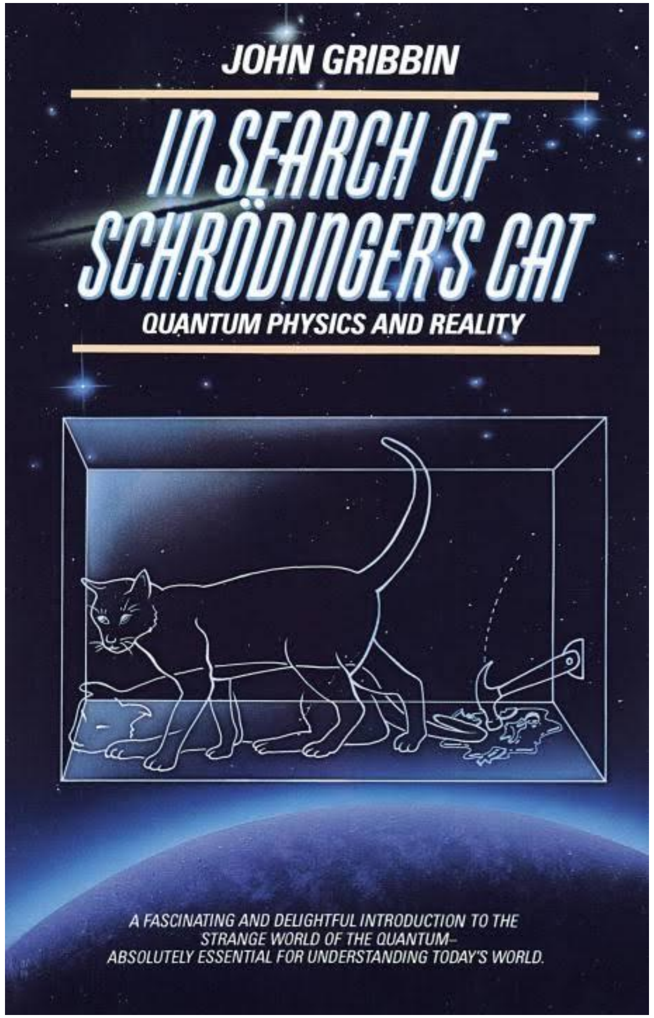
- 按你的考试范围做一份**针对性速记提纲/公式表**。
- 出一套**模拟题**（选择/计算/简答）并给详细解析。
- 讲清**易错点**（如酸碱滴定、气体定律、热化学、平衡与勒夏特列、溶解度积、电化学等）。
- 逐题训练你的**计算步骤与单位换算**，教你常见**估算与检查技巧**。

你把考试范围、老师强调的章节（或往年题型）发给我，我现在就给你做一套高效复习包。

- **需要注意内容的严谨性，学术诚信等**

https://www.bilibili.com/bangumi/play/ep377412

Other online resources



In Search of Schrödinger's Cat: Quantum Physics and Reality

- 需要注意内容的严谨性

Class syllabus

Basics
- Formula
- Classical atom & molecule

Atoms
- Quantum mech.
- Hydrogen atom
- Periodic table

Molecules
- Geometry
- Bonding
- Coordination
- Gas, solid, liquid

Light & Electricity
- Photochemistry
- Electrochemistry

Chapters (Principles of Modern Chemistry, 7th)

Basics	Atoms	Molecules	Light & Electricity
1,2,3	4,5	6,7,8,9,10	17,20

九月				十月					十一月				十二月				一月		
1	8	15	22	29	6 中秋节	13	20	27	3	10	17	24	1	8	15	22	29	5	12
2	9	16	23	30	7	14	21	28	4	11	18	25	2	9	16	23	30	6	13
3	10	17	24	1 国庆节	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14
4	11	18	25	2	9	16	23	30	6	13	20	27	4	11	18	25	1 元旦	8	15
5	12	19	26	3	10	17	24	31	7	14	21	28	5	12	19	26	2	9	16
6	13	20	27	4	11	18	25	1	8	15	22	29	6	13	20	27	3	10	17
7	14	21	28	5	12	19	26	2	9	16	23	30	7	14	21	28	4	11	18
7	8	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
假		秋学期																	

Week	Monday	Wednesday	Friday
1	subject to change	Sept 17 Overview & Introduction	Sept 19 The atom in modern chemistry
2	Sept 22 Recitation		Sept 26 Chemical formulas, equations, and reaction yields Quiz 1
3	Sept 29 PS1 due, Recitation	Oct 1 holiday	Oct 3 holiday
4	Oct 6 holiday		Oct 10 Atomic shells and classical models of chemical Quiz 2
5	Oct 13 Recitation	Oct 15 Atomic shells and classical models of chemical	Oct 17 Introduction to quantum mechanics-1 Quiz 3
6	Oct 20 PS2 due, Recitation		Oct 24 Introduction to quantum mechanics-2 Review 1 Quiz 4
7	Oct 27 Recitation	Oct 29 Midterm 1	Oct 31 Quantum mechanics and atomic structure-1 Quiz 5
8	Nov 3 PS3 due, Recitation		Nov 7 Quantum mechanics and atomic structure-2 Quiz 6
9	Nov 10 Recitation	Nov 12 Quantum mechanics and molecular structure-1	Nov 14 Quantum mechanics and molecular structure-2 Quiz 7
10	Nov 17 PS4 due, Recitation		Nov 21 Quantum mechanics and molecular structure-3 Quiz 8
11	Nov 24 Recitation	Nov 26 Quantum mechanics and molecular structure-4	Nov 28 Quantum mechanics and molecular structure-5 Quiz 9
12	Dec 1 PS5 due, Recitation		Dec 5 Bonding in organic molecules Review 2 Quiz 10
13	Dec 8 Recitation	Dec 10 Midterm 2	Dec 12 Bonding in transition metal compounds and coordination complexes-1 Quiz 11
14	Dec 15 PS6 due, Recitation		Dec 19 Bonding in transition metal compounds and coordination complexes-2 Quiz 12
15	Dec 22 Recitation	Dec 24 Gas, solids, liquid, and phase transitions	Dec 26 Rare earths Quiz 13
16	Dec 29 PS7 due, Recitation		Jan 2 Spectroscopy Review 3 Quiz 14

Class syllabus

Quiz x13: 26%

- 随堂测验一共14次，计分时舍弃一个最低分，每次15-25分钟

Problem set x7: 28%

- 周五课后发布
- 下一个周一习题课前（下午三点）提交至bb系统，每次总分为 4 分；周二上午10点前补交，每次总分为 3 分（即实际成绩乘以 3/4）；学期结束前补交，每次总分为 2 分
- 学生之间可以就课后习题进行讨论，但不允许相互抄袭，建议先充分讨论然后再开始单独动笔解答。任课老师和助教有权将疑似抄袭的作业提交学校的有关委员会进行鉴定

Project: 6%

Mid-term exam x2: 20%

- 因故无法参加期中考试的学生需按照学校规定请假

Final exam x1: 20%

- 17-18周进行

Q & A

- **Where to get the textbooks?**
- **How to read the textbooks?**
- **Can I read the textbooks in Chinese only?**
- **Other questions?**

Lecture 1-2: introduction

Jin Xie (谢瑨)

2025/09/17

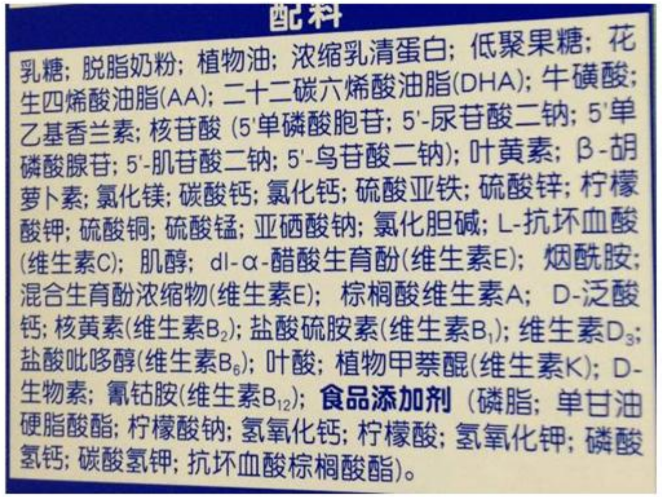
Major points of today's lecture

1. Discuss class syllabus

2. Establish the context of general chemistry

What to learn in this course?

- **Understand basic chemistry principles**
- **Handle vast amounts of (scattered) information**
- **Identify the key points**
- **Make reasonable judgments**
- **Lifelong learning**



某品牌婴幼儿配方奶粉配料表

带何种口罩？什么是气溶胶？84消毒液是什么？怎么配置75%浓度酒精溶液？

在家怎么做可乐鸡翅呢？用什么可乐？什么是分子料理？

nature chemistry

ARTICLES
PUBLISHED ONLINE: 2014

A comprehensive overview of chemical-free consumer products

Alexander F. G. Goldberg[1] and CJ Chemjobber[2]*

Manufacturers of consumer products, in particular edibles and cosmetics, have broadly employed the term 'Chemical free' in marketing campaigns and on product labels. Such characterization is often incorrectly used to imply — and interpreted to mean — that the product in question is healthy, derived from natural sources, or otherwise free from synthetic components. We have examined and subjected to rudimentary analysis an exhaustive number of such products, including but not limited to lotions and cosmetics, herbal supplements, household cleaners, food items, and beverages. Herein are described all those consumer products, to our knowledge, that are appropriately labelled as 'Chemical free'.

[1]Department of Organic Chemistry, Weizmann Institute of Science, Rehovot 76100, Israel, [2]3170 Road 40 1/2, Shell, WY 82441, USA.
*e-mail: chemjobber@gmail.com

References

1. 'Chemical-free' sunscreen: http://campl.us/bmnl
2. 'Chemical-free' chemistry set: http://sciencegeist.net/my-chemically-fueled-life/
3. 'Chemical-free' bassinets: http://www.nytimes.com/2012/03/15/garden/going-to-extreme-lengths-to-purge-household-toxins.html?pagewanted=1&_r=1&
4. 'Chemical-free' eggs: http://justlikecooking.blogspot.com/2012/07/chemophobia-vacation-style.html

Acknowledgments

CJC thanks Carmen Drahl for pioneering this important topic in the modern chemistry blogosphere. A.F.G.G. thanks the Azrieli Foundation for an Azrieli Postdoctoral Fellowship.

Author contributions

Both authors contributed equally to the main text.

Additional information

Correspondence should be addressed to chemjobber@gmail.com, including requests for reprints and permission information.

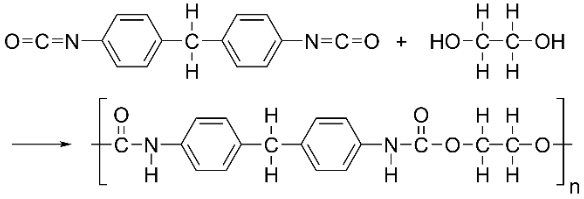
Competing financial interests

The authors declare no competing financial interests, though would have short-sold 'Rubber Ducky Sunscreen' on principle if it was publicly traded.

Outline

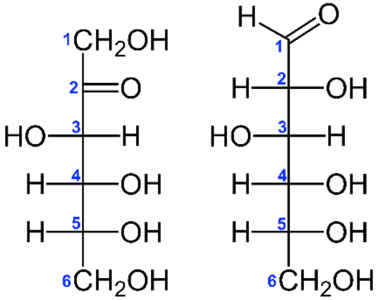
- Why is chemistry central to our life?
- What does chemistry study?
- What has chemistry achieved and what hasn’t?
- What do we benefit from learning chemistry?

衣



食

High-fructose corn syrup



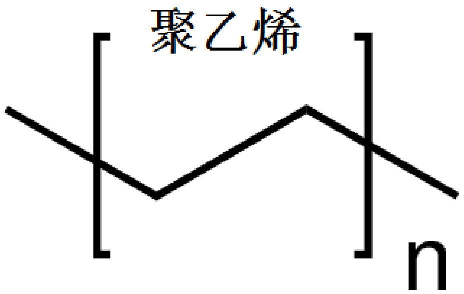
住

Calcite: $CaCO_3$
Dolomite: $CaMg(CO_3)_2$

行

Cathode: NCA ($LiNi_xCo_yAl_zO_2$)
Anode: graphite (C)

衣



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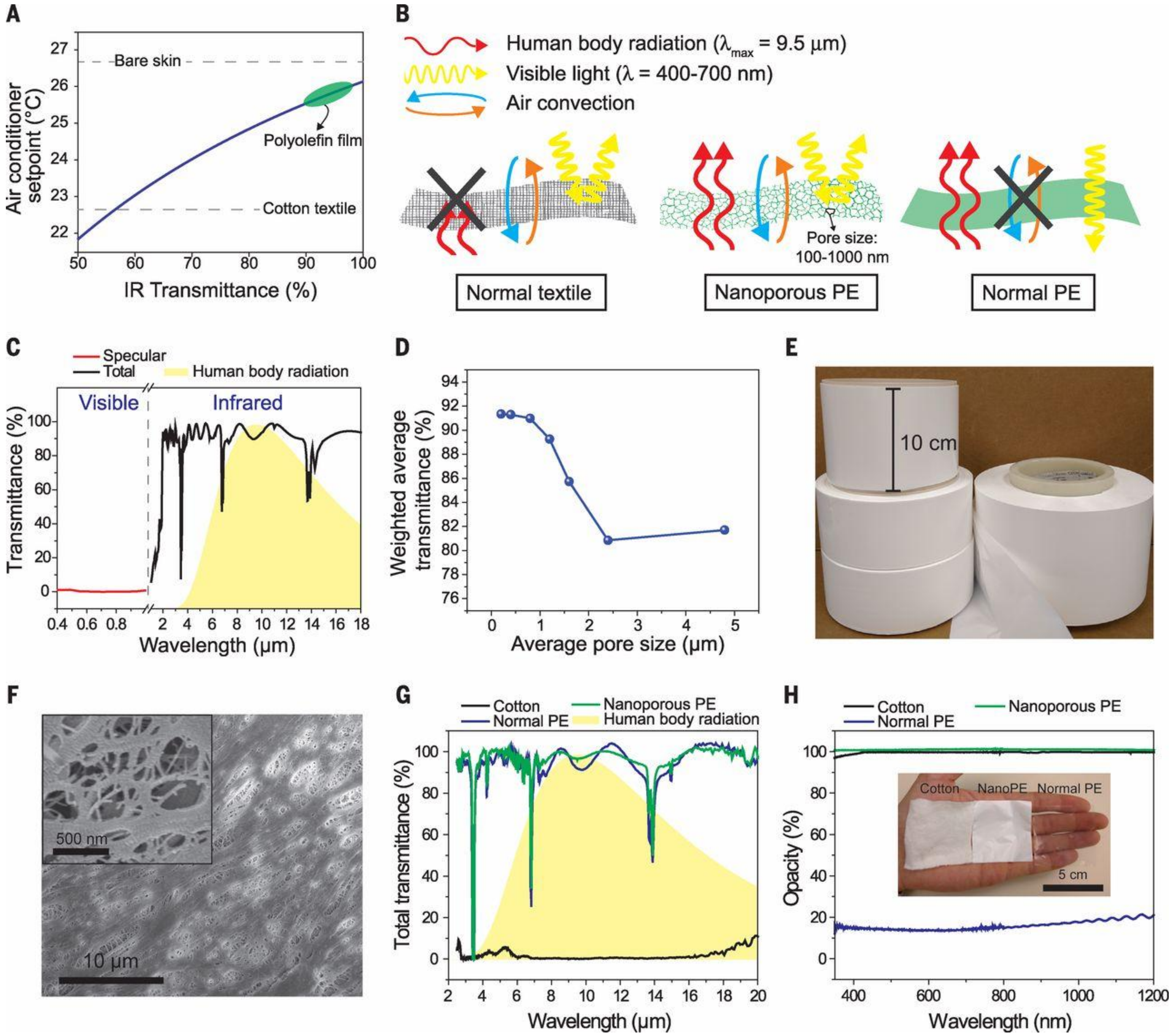
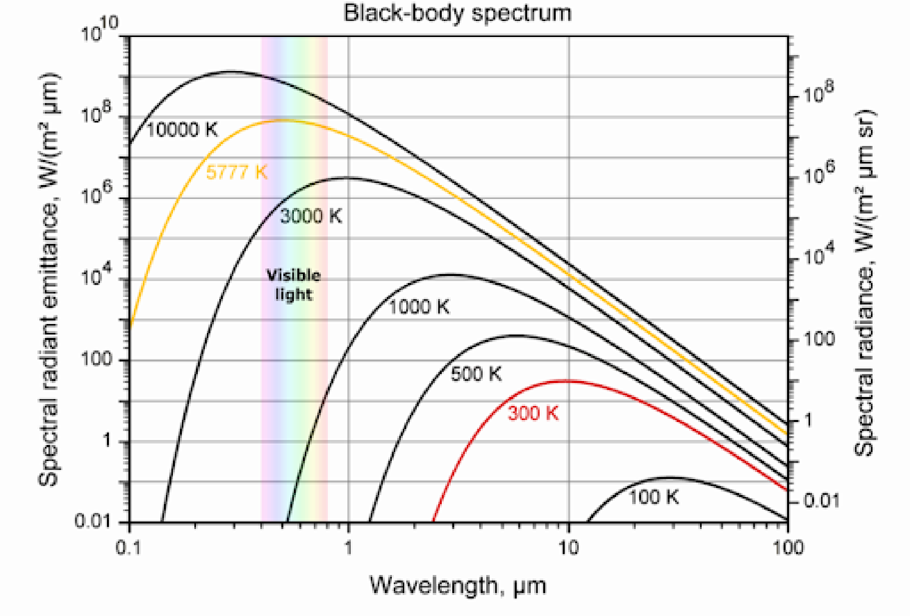




P.-C. Hsu, Science 2016, 353, 1019-1023.

食

Carbonated water
High fructose corn syrup
Caramel color
Phosphoric acid
Natural flavors
Caffeine

Carbonated water
Cane sugar
Caramel color
Natural flavors
Phosphoric acid
Potassium benzoate
Caffeine
Stevia leaf extract (steviol glycosides)

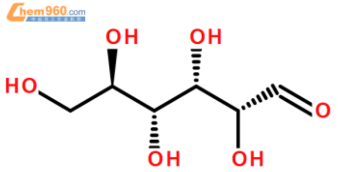
Carbonated water
Caramel color
Aspartame
Phosphoric acid
Potassium benzoate
Natural flavors
Citric acid
Caffeine

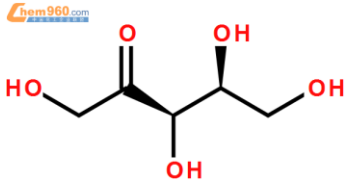
Carbonated water
Caramel color
Phosphoric acid
Aspartame
Potassium benzoate
Natural flavors
Potassium citrate
Acesulfame potassium
Caffeine

怎么做可乐鸡翅呢？应该选哪个可乐？

食

HIGH FRUCTOSE CORN SYRUP
高果糖浆（葡萄糖，果糖）



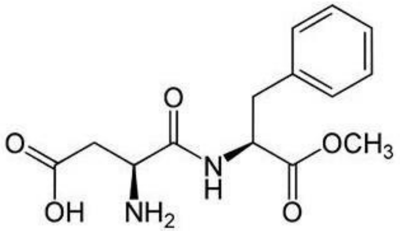


褐变现象

- **美拉德反应**

美拉德反应亦称非酶棕色化反应，是广泛存在于食品工业的一种非酶褐变。是羰基化合物（还原糖类）和氨基化合物（氨基酸和蛋白质）间的反应，经过复杂的历程最终生成棕色甚至是黑色的大分子物质类黑精或称拟黑素，故又称羰胺反应（1912年法国化学家L.C.Maillard提出）。

ASPARTAME
阿斯巴甜



- **焦糖化反应**

焦糖化反应是糖类尤其是单糖在没有氨基化合物存在的情况下，加热到熔点以上的高温（一般是140-170℃以上）时，因糖发生脱水与降解，也会发生褐变反应。

阿斯巴甜危害及风险评估结果发布

2023年7月14日 | 联合新闻发布 | 日内瓦

国际癌症研究机构、世界卫生组织（世卫组织）和粮食及农业组织（粮农组织）食品添加剂联合专家委员会（下称联合专家委员会）今天发布了无糖甜味剂阿斯巴甜对健康影响的评估报告。国际癌症研究机构援引对人类致癌性的"有限证据"，将阿斯巴甜归为可能对人类致癌之列（国际癌症研究机构第2B组），联合专家委员会重申其每日允许摄入量为每公斤体重40毫克。

阿斯巴甜是一种人造（化学）甜味剂，自20世纪80年代以来广泛用于各种食品和饮料制品，包括减肥饮料、口香糖、明胶、冰淇淋、酸奶等乳制品、早餐麦片、牙膏和止咳药水和维生素咀嚼片等药物。

"癌症是全球主要死因之一。每年，每6人中就有1人死于癌症。正在不断扩大科学探索，评估癌症的可能诱发或促进因素，以期减少患病数和人类死亡，"世卫组织营养和食品安全司司长Francesco Branca博士说。"对阿斯巴甜的评估表明，虽然在常用量方面安全性不是主要问题，但已对潜在影响作了描述，需要通过更多更好的研究来进行调查。"

两个机构的审查既独立又相互补充，目的是评估与食用阿斯巴甜相关的潜在致癌危害和其他健康风险。这是国际癌症研究机构首次对阿斯巴甜进行评估，也是联合专家委员会对其开展的第三次评估。

经过审查现有的科学文献，两方面的评估都指出癌症（和其他健康影响）方面的现有证据存在局限性。

国际癌症研究机构根据人类癌症（特别是肝细胞癌这种肝癌）方面的有限证据，将阿斯巴甜归类为可能对人类致癌（第2B组）。在实验动物中发现的致癌证据也有限，与导致癌症的可能机理有关的证据同样有限。

联合专家委员会得出结论，所评估的数据表明没有充分理由改变以往确定的阿斯巴甜每公斤体重0-40毫克这一每日允许摄入量。因此，委员会重申，人们可在这个每日限量内放心食用。例如，假设没有其他方面的食物摄入，一罐含有200或300毫克阿斯巴甜的减肥软饮料，一位体重70公斤的成人每天要饮用9-14罐以上才会超过每日允许摄入量。

International Agency for Research on Cancer

World Health Organization

IARC MONOGRAPHS HAZARD CLASSIFICATION

IARC Group	Level of certainty that a substance can cause cancer (typical examples of evidence leading to each group)	Substances evaluated
GROUP 1	**CARCINOGENIC TO HUMANS** *Sufficient* evidence for cancer in humans.	Tobacco smoking, solar radiation, consumption of alcoholic beverages, ionizing radiation
GROUP 2A	**PROBABLY CARCINOGENIC TO HUMANS** *Limited* evidence for cancer in humans. *Sufficient* evidence in experimental animals.	Emissions from high-temperature frying, DDT, consumption of red meat, night shift work
GROUP 2B	**POSSIBLY CARCINOGENIC TO HUMANS** *Limited* evidence in humans. Less than *sufficient* evidence in experimental animals.	Gasoline engine exhaust, occupational exposure as a hairdresser or barber, lead
GROUP 3	**NOT CLASSIFIABLE AS TO ITS CARCINOGENICITY TO HUMANS** *Inadequate* evidence in humans. *Inadequate* evidence in experimental animals.	Coffee drinking, crude oil, mercury, paracetamol

Higher level of certainty

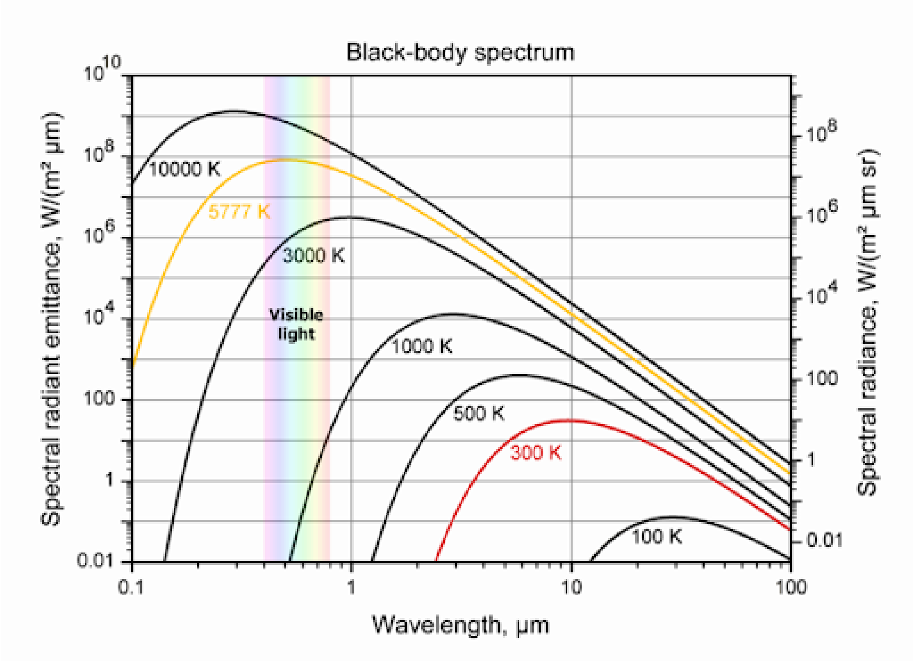
Lower level of certainty

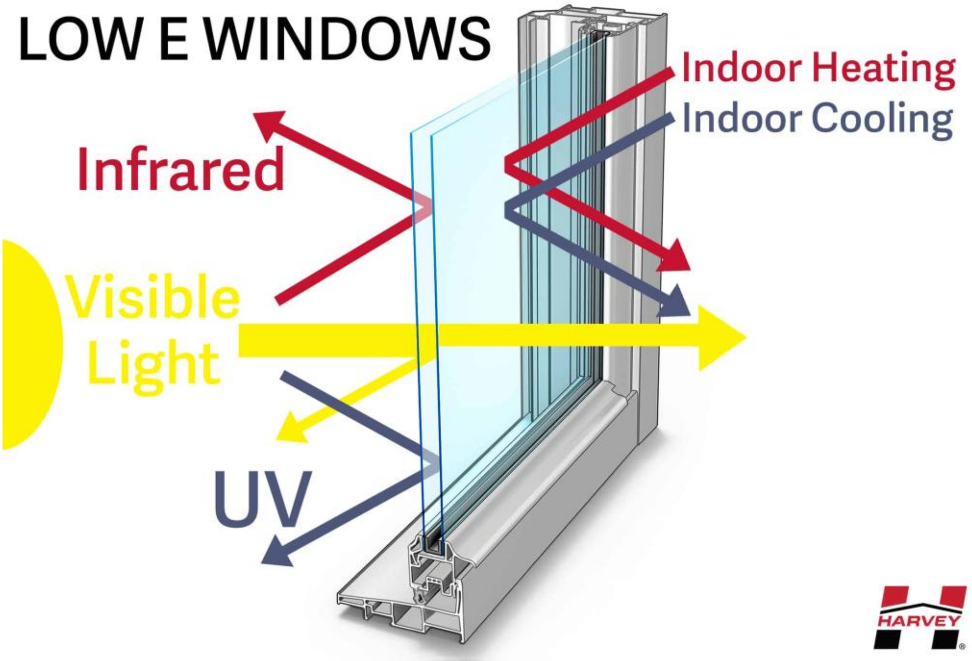
The classification indicates the level of certainty that a substance can cause cancer (*hazard identification*)

This classification does not indicate the level of risk associated with exposure (*risk assessment*)

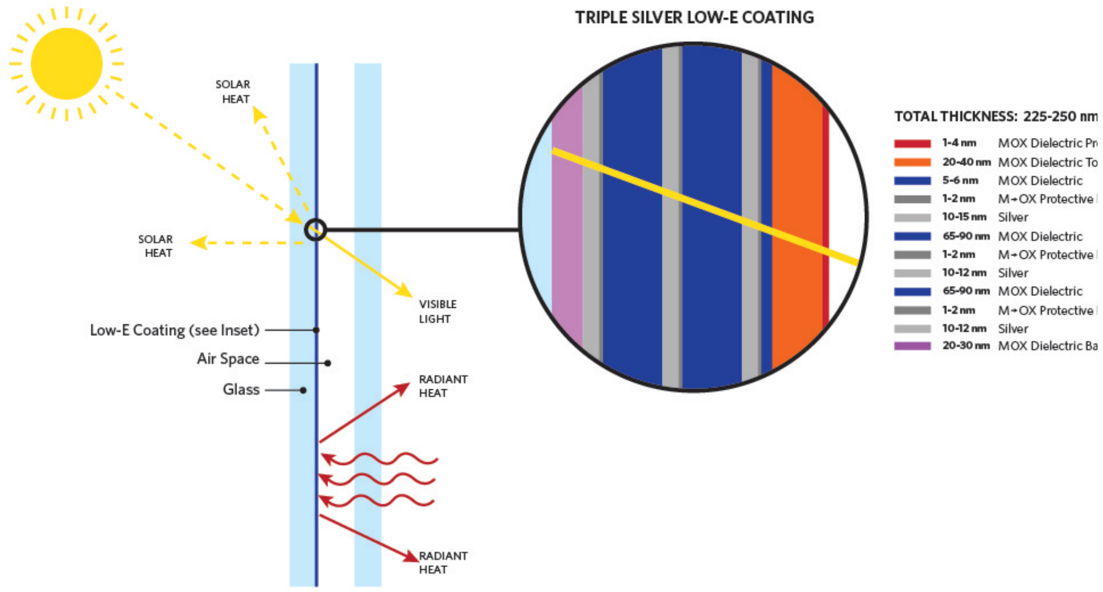
June 2023

住

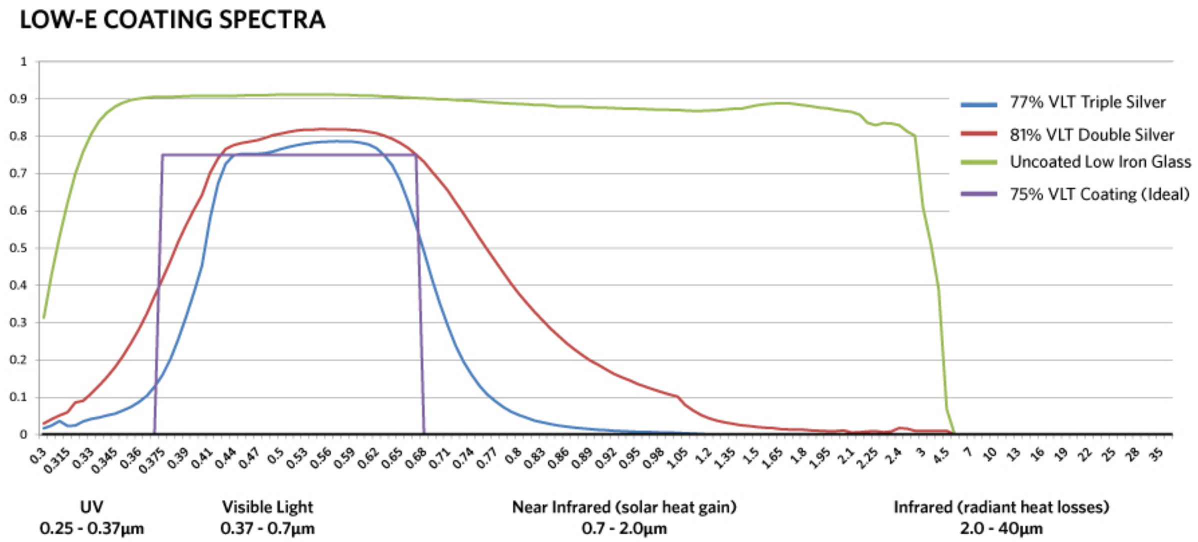

Black-body spectrum
Spectral radiant emittance, W/(m² μm)
Spectral radiance, W/(m² μm sr)
Wavelength, μm
10000 K
5777 K
3000 K
1000 K
500 K
300 K
100 K
Visible light
0.1
1
10
100
0.01


LOW E WINDOWS
Indoor Heating
Indoor Cooling
Infrared
Visible Light
UV
HARVEY

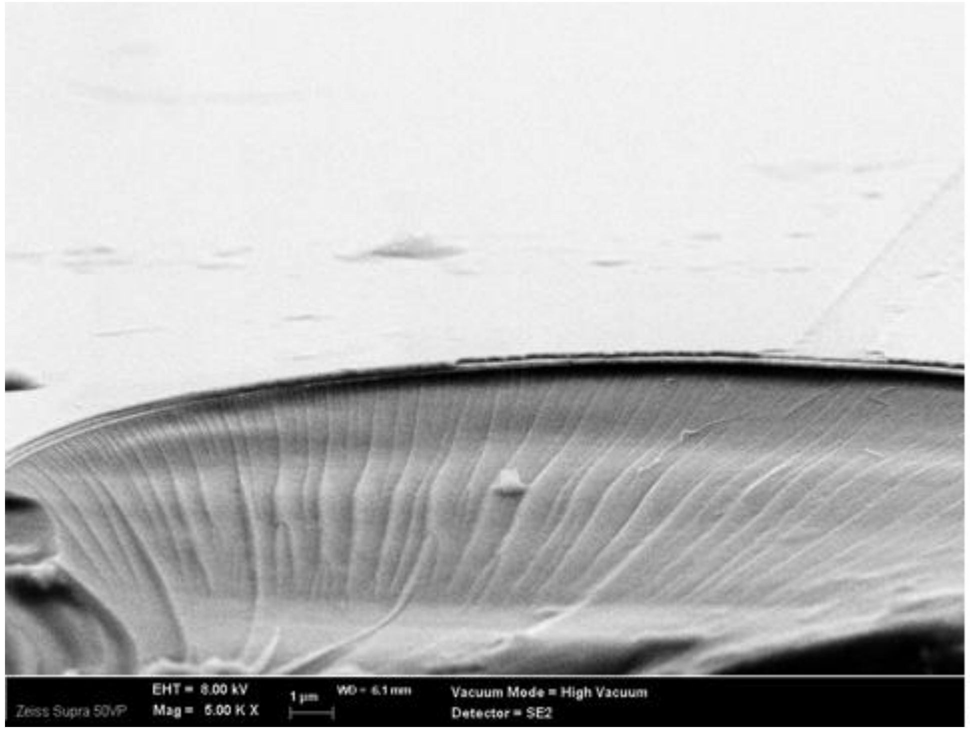
住



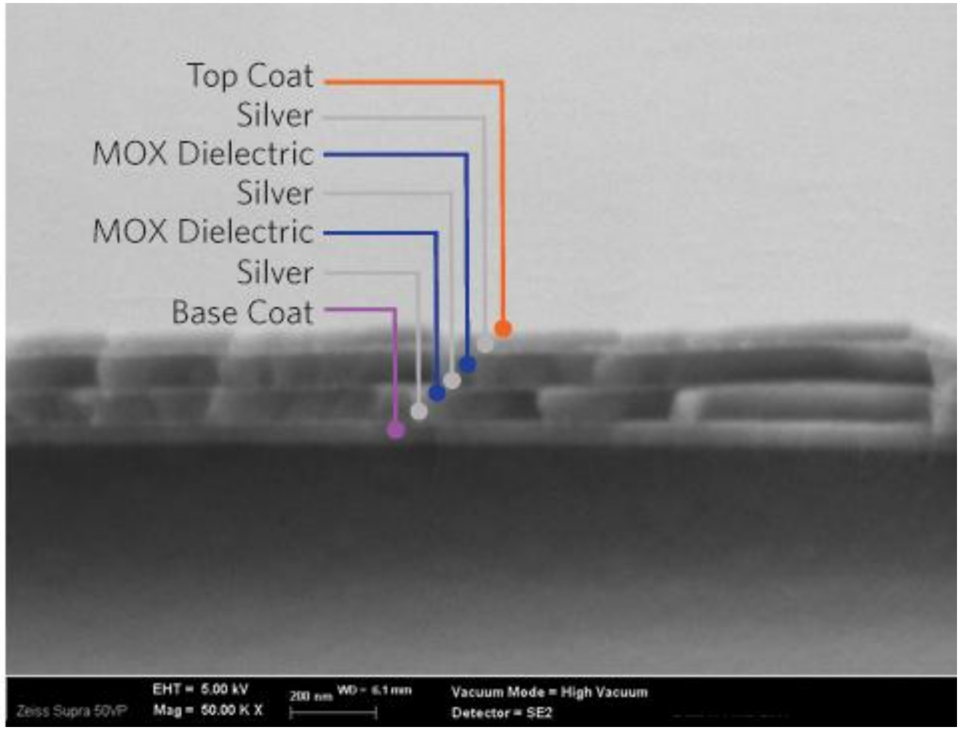
TRIPLE SILVER LOW-E COATING
SOLAR HEAT
SOLAR HEAT
VISIBLE LIGHT
RADIANT HEAT
RADIANT HEAT
Low-E Coating (see Inset)
Air Space
Glass
TOTAL THICKNESS: 225-250 nm
1-4 nm MOX Dielectric Pr
20-40 nm MOX Dielectric To
5-6 nm MOX Dielectric
1-2 nm M+OX Protective
10-15 nm Silver
65-90 nm MOX Dielectric
1-2 nm M+OX Protective
10-12 nm Silver
65-90 nm MOX Dielectric
1-2 nm M+OX Protective
10-12 nm Silver
20-30 nm MOX Dielectric Ba



LOW-E COATING SPECTRA
77% VLT Triple Silver
81% VLT Double Silver
Uncoated Low Iron Glass
75% VLT Coating (Ideal)
UV
0.25 - 0.37µm
Visible Light
0.37 - 0.7µm
Near Infrared (solar heat gain)
0.7 - 2.0µm
Infrared (radiant heat losses)
2.0 - 40µm



EHT = 8.00 kV
Mag = 5.00 K X
WD = 6.1 mm
Vacuum Mode = High Vacuum
Detector = SE2
Zeiss Supra 50VP



Top Coat
Silver
MOX Dielectric
Silver
MOX Dielectric
Silver
Base Coat
EHT = 5.00 kV
Mag = 50.00 K X
WD = 6.1 mm
Vacuum Mode = High Vacuum
Detector = SE2
Zeiss Supra 50VP

住

13-18 AC
“Damascus” saber blades

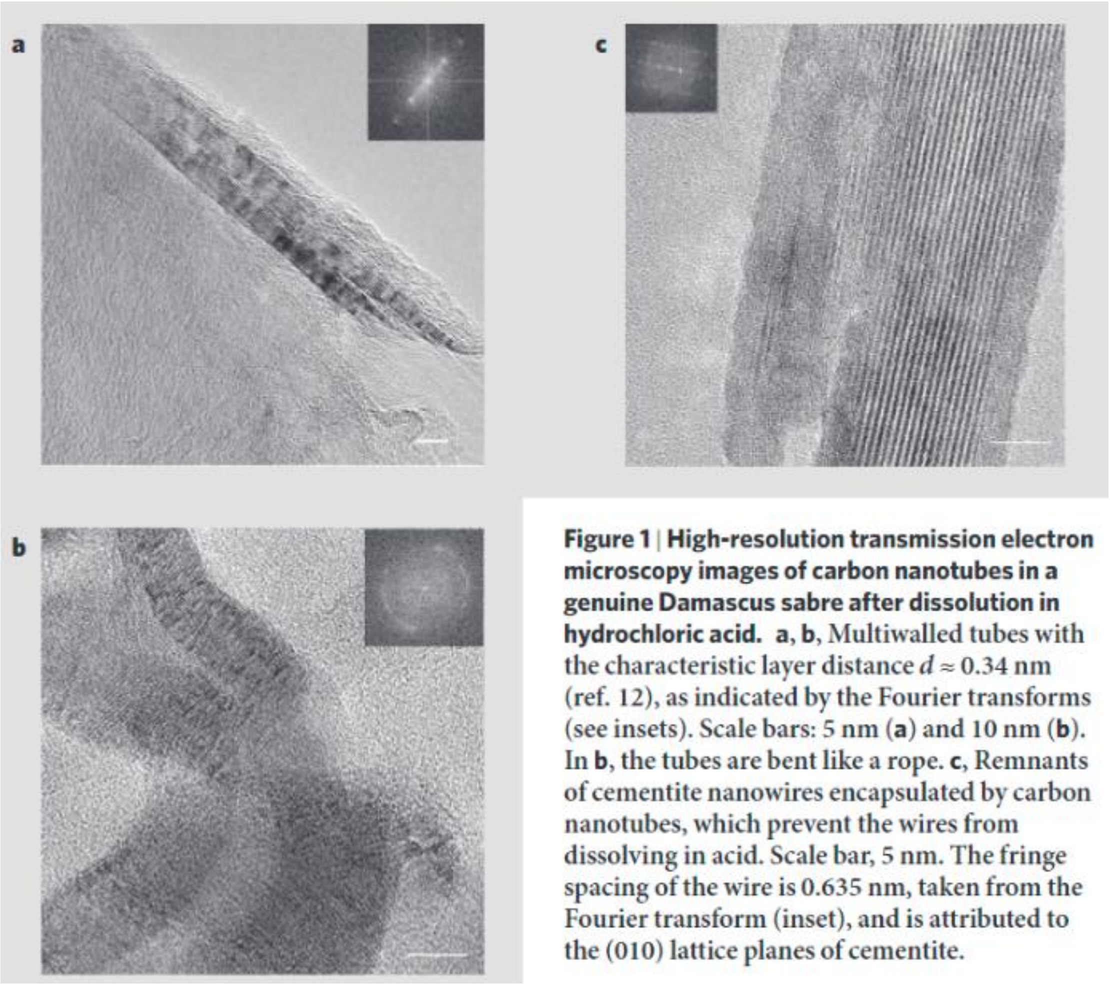


Figure 1 | High-resolution transmission electron microscopy images of carbon nanotubes in a genuine Damascus sabre after dissolution in hydrochloric acid. **a, b,** Multiwalled tubes with the characteristic layer distance $d \approx 0.34$ nm (ref. 12), as indicated by the Fourier transforms (see insets). Scale bars: 5 nm (**a**) and 10 nm (**b**). In **b**, the tubes are bent like a rope. **c,** Remnants of cementite nanowires encapsulated by carbon nanotubes, which prevent the wires from dissolving in acid. Scale bar, 5 nm. The fringe spacing of the wire is 0.635 nm, taken from the Fourier transform (inset), and is attributed to the (010) lattice planes of cementite.

(carbon nanotubes and Fe_3C nanowires)

https://www.nature.com/articles/444286a

Battery revolution: electrical vehicles

A healthy horse
Bioenergy (ATP)

1 horsepower=550lb×1ft/1s;
0-60 mph: ?

Mustang Shelby GT500
Chemical energy (gasoline)

760 horsepower;
0-60 mph: 3.3s

Tesla Model S Plaid
Electrical energy (Lithium ion battery)

1100+ horsepower;
0-60 mph: <2.0s

Technology Revolution: Electrical Grid




Technology revolution: electrical grid

Usable Capacity
13.5 kWh

Depth of Discharge
100%

Efficiency
90% round-trip

Power
7kW peak / 5kW continuous

Supported Applications
Solar self-consumption
Back-up power
Time-Based control
Off-grid capabilities

Warranty
10 years

Scalable
Up to 10 Powerwalls

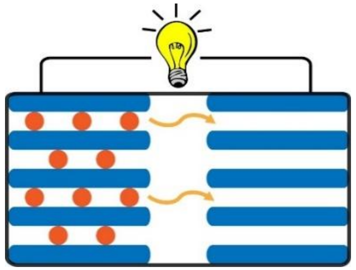
Operating Temperature
-4°F to 122°F / -20°C to 50°C

Dimensions
L x W x D: 45.3" x 29.6" x 5.75"
(1150 mm x 753 mm x 147 mm)

Weight
251.3 lbs / 114 kg

Installation
Floor or wall mounted
Indoor or outdoor

Certification
North American and International Standards
Grid code compliant



Lithium-ion Battery

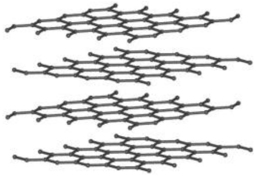
intercalation

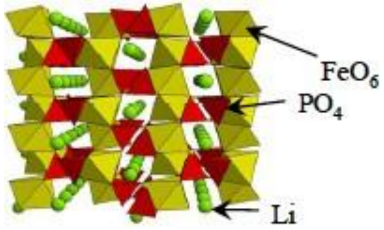
$$C_6 + LiCoO_2 \rightleftarrows Li_xC_6 + Li_{(1-x)}CoO_2$$

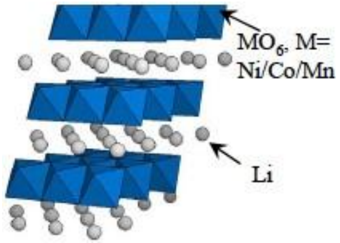
Graphite (2D) **(~370 mAh/g)**

$LiFePO_4$ (1D) **(~170mAh/g)**

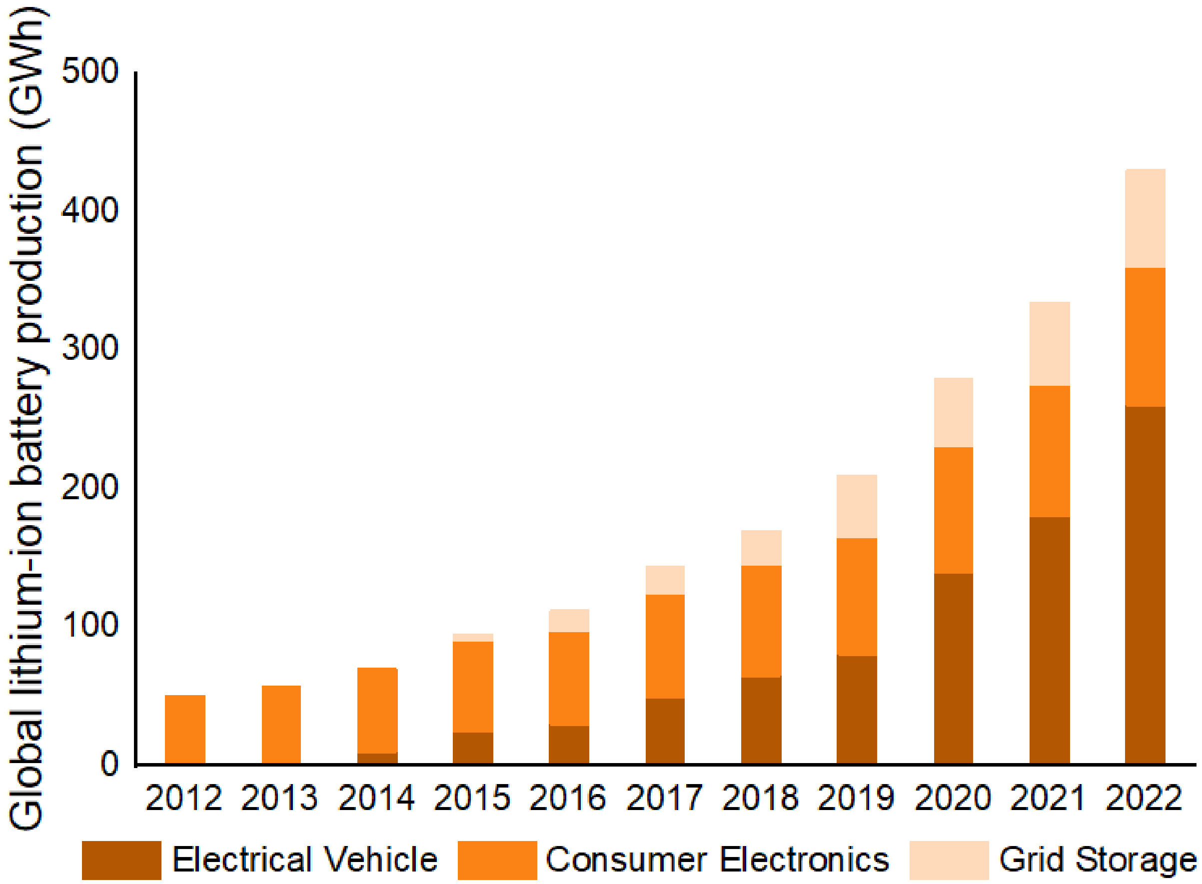
$LiMO_2$(2D) **(~140 mAh/g)**



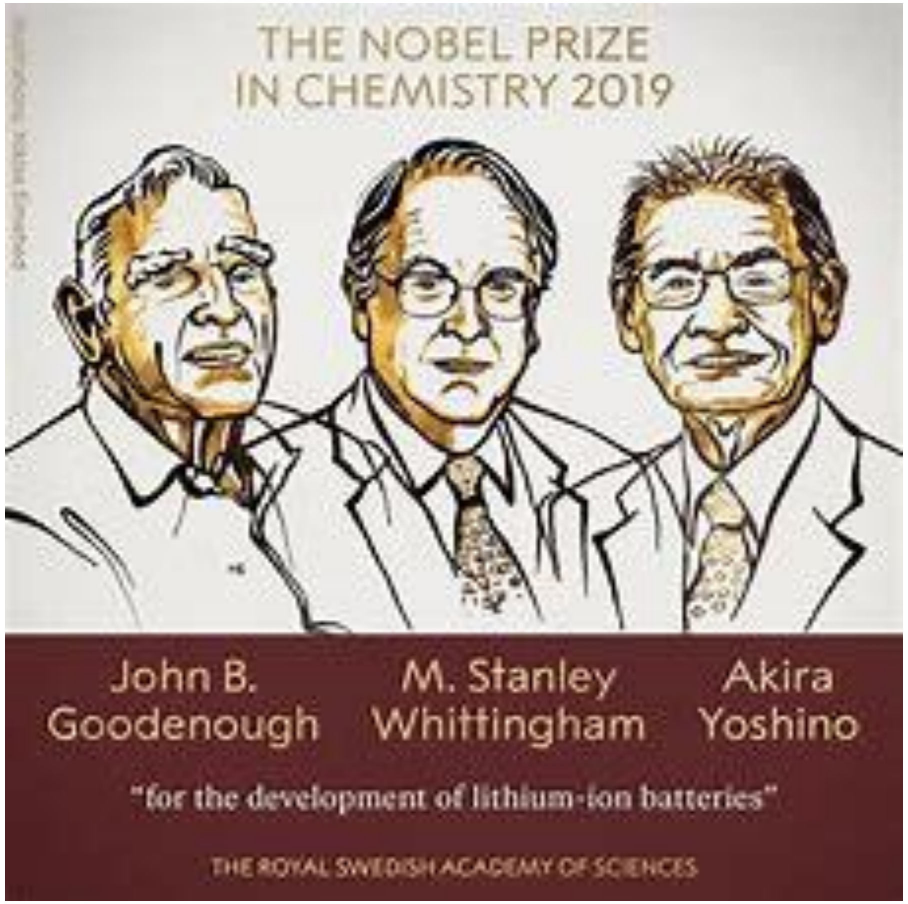




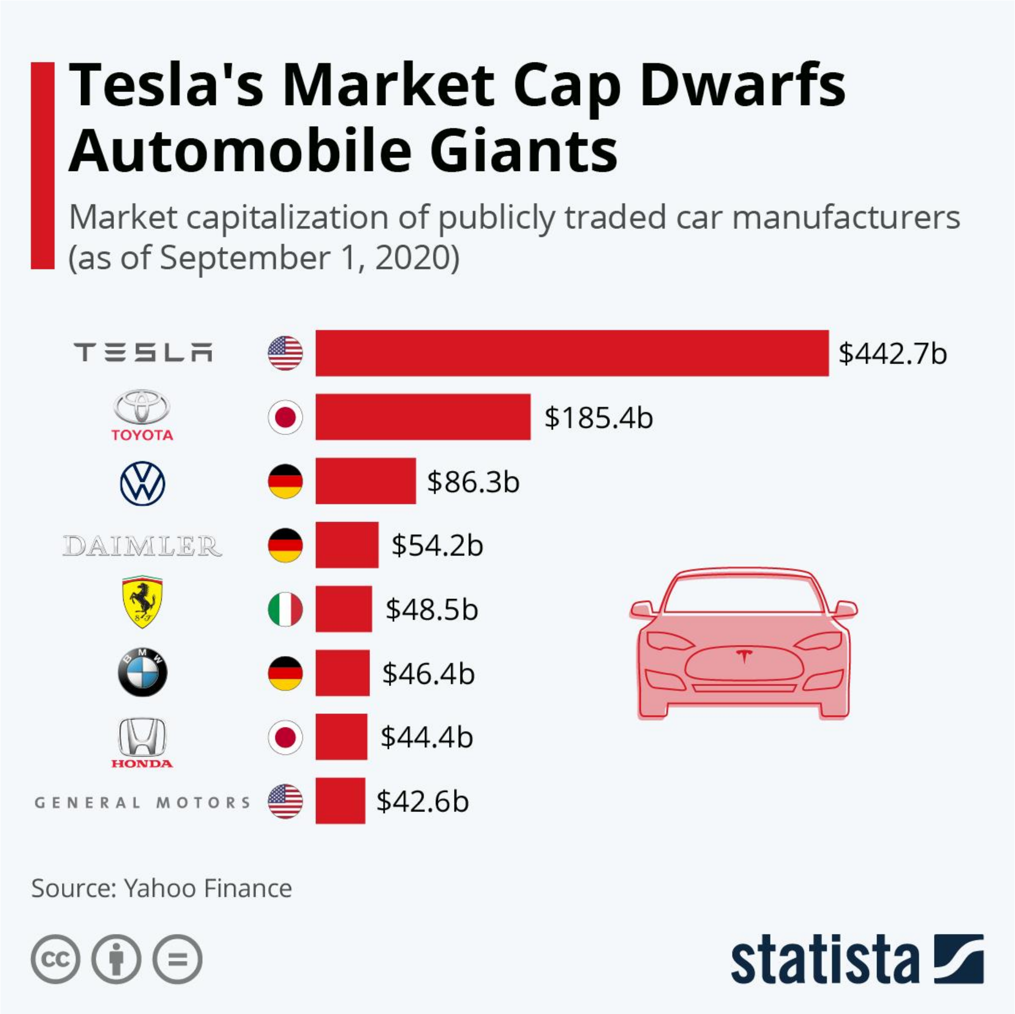
Development of lithium ion batteries

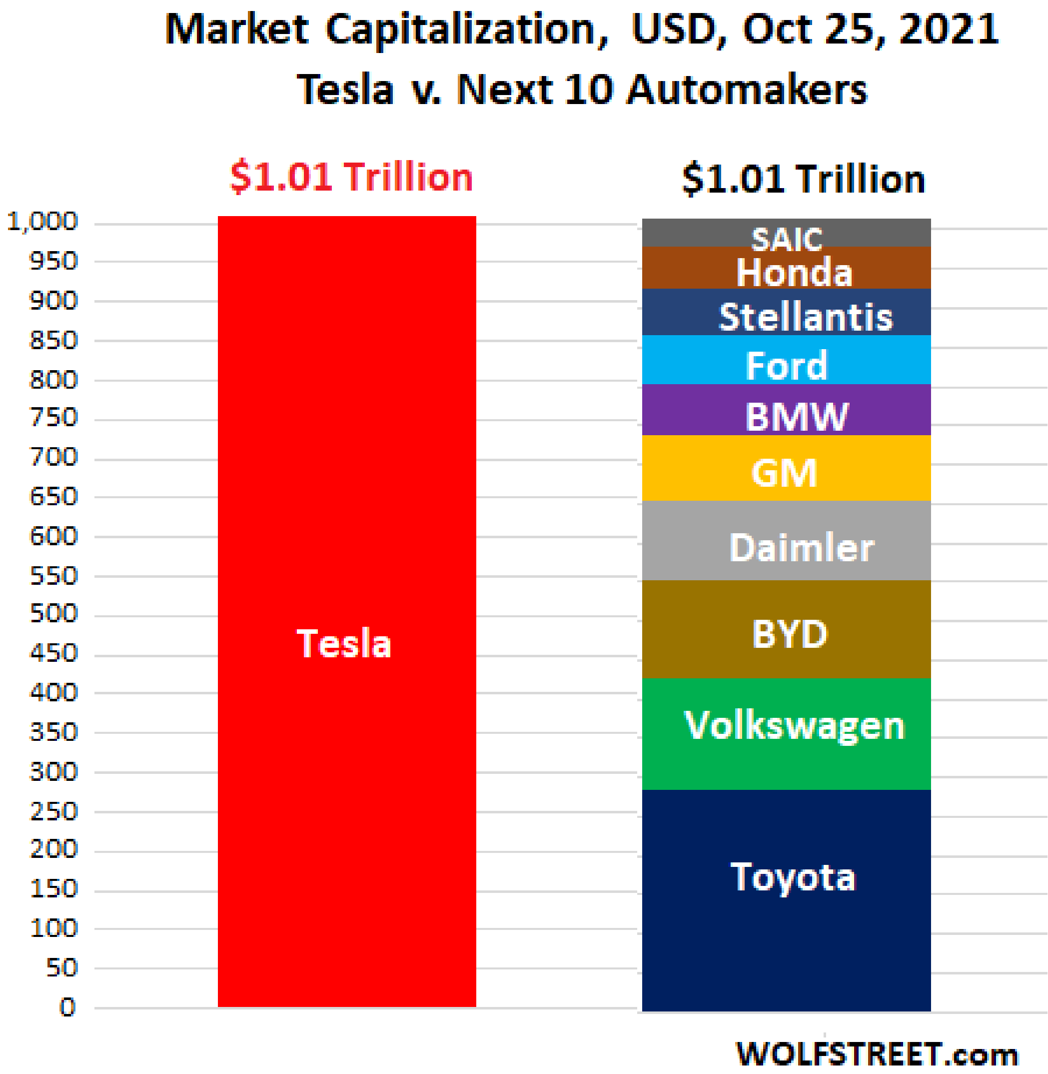


(data source: EVTank)

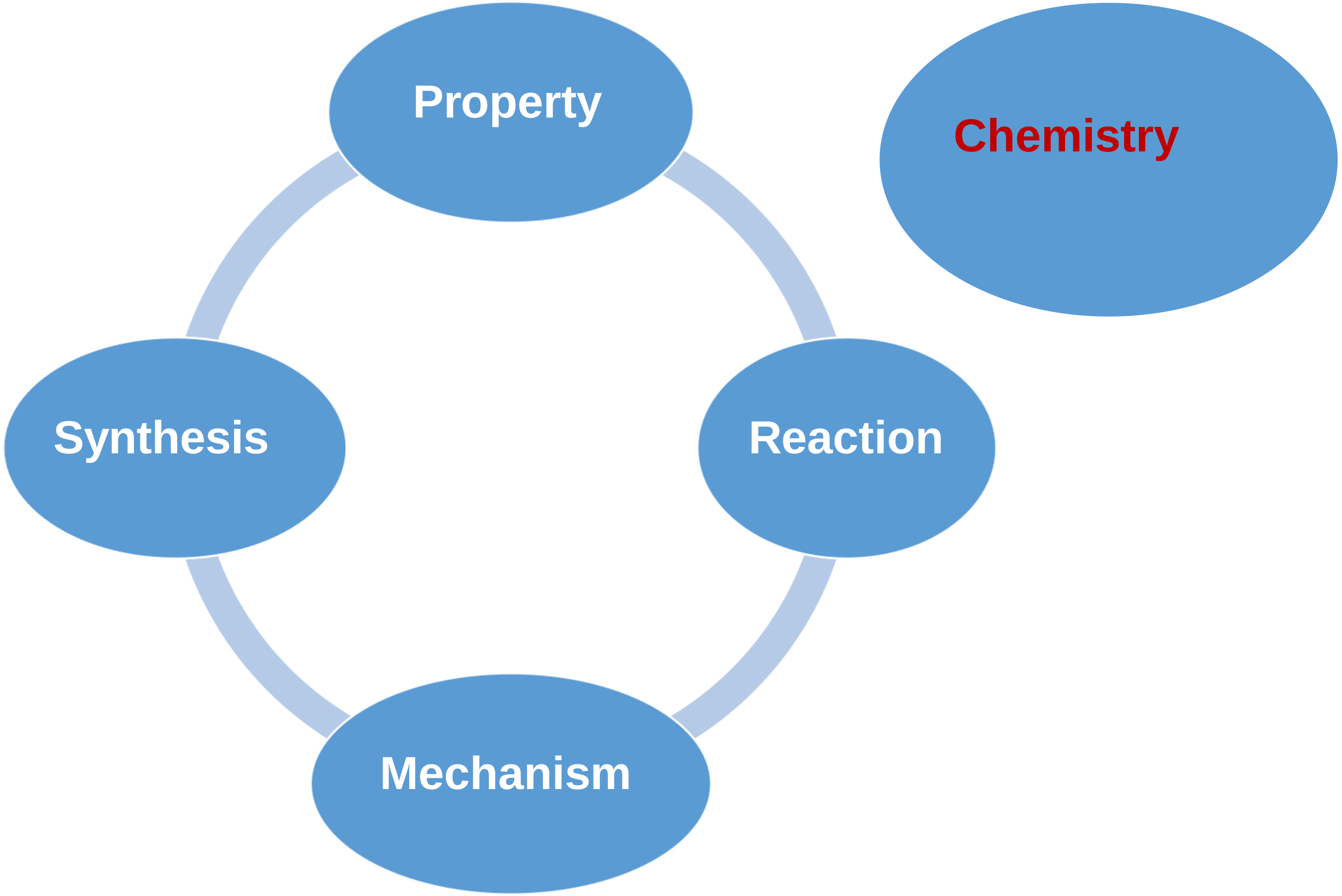


Development of lithium ion batteries





Any more examples?



Outline

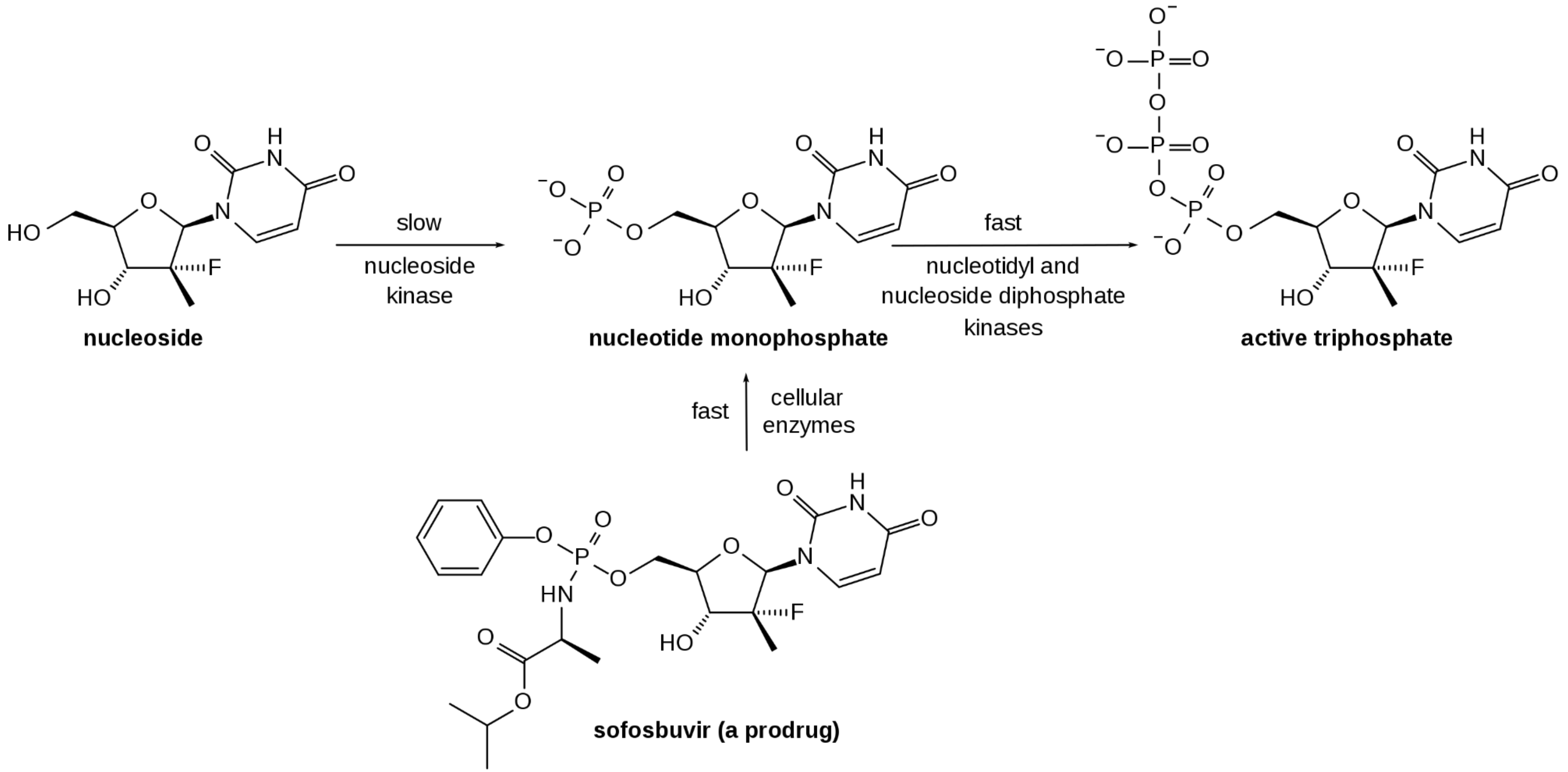
- Why is chemistry central to our life?
- What does chemistry study?
- What has chemistry achieved and what hasn’t?
- What do we benefit from learning chemistry?

What does chemistry study?

- Create entirely new substances designed to have specific properties
- Properties of substances
- The reactions that transform substances into other substances
- Understanding how and why specific chemical reactions occur
- Tailor the properties of existing substances to meet particular needs

Creating new substances

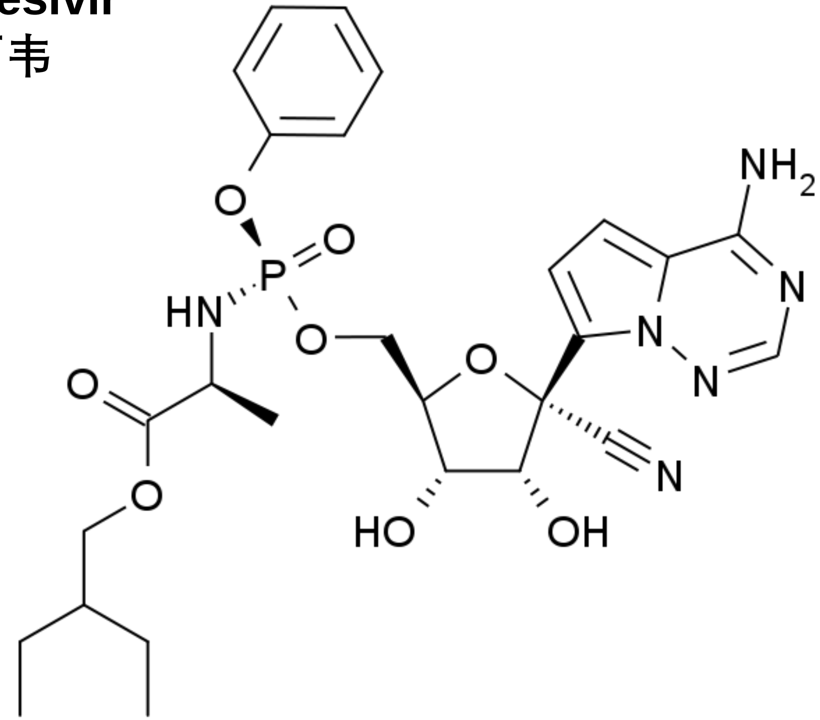
Sofosbuvir
索非布韦



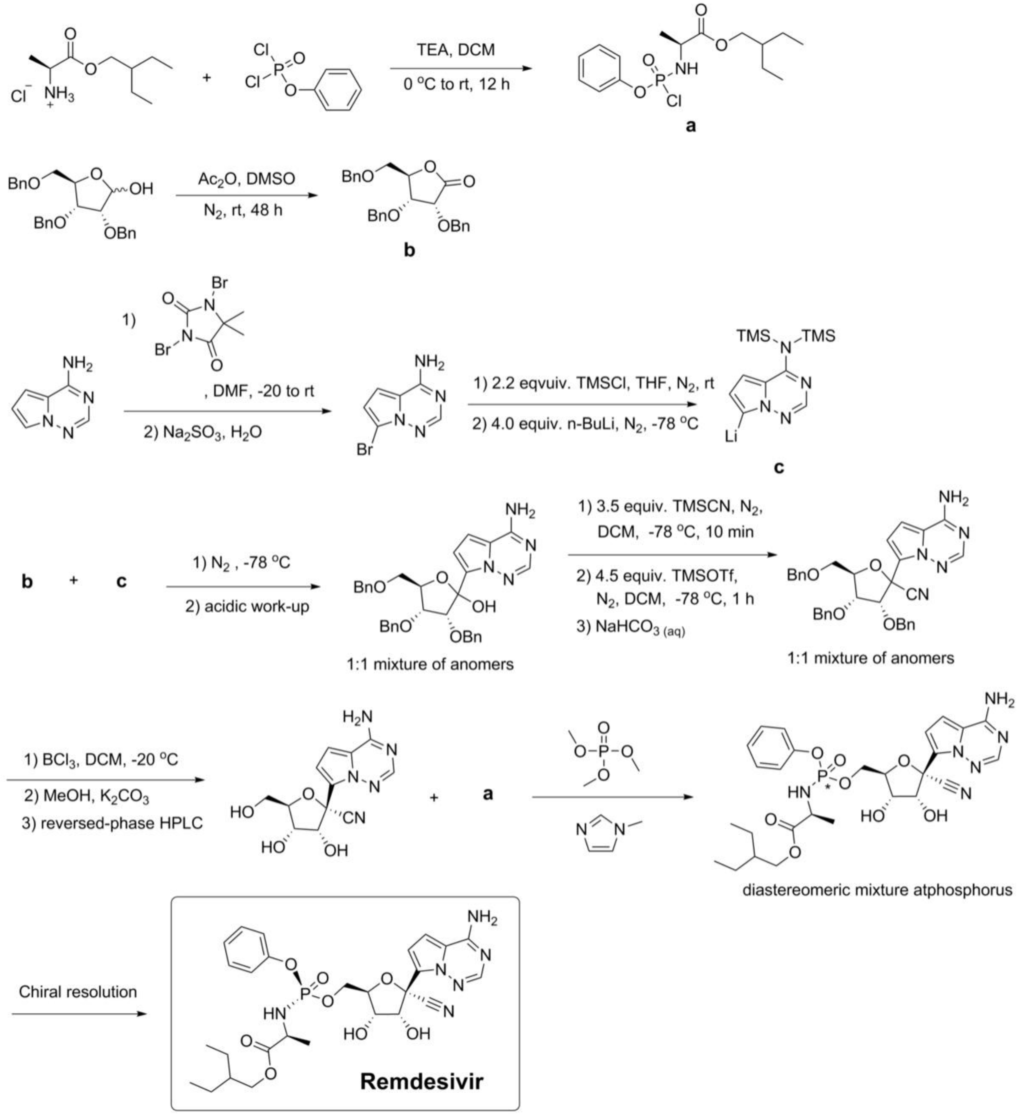
Formula $C_{22}H_{29}FN_3O_9P$
Molar mass 529.453 g/mol g·mol⁻¹

Creating new substances

Remdesivir
瑞德西韦



Formula $C_{27}H_{35}N_6O_8P$
Molar mass 602.585 g·mol^{-1}



On the origin of the blind chicken molecule

A curious chemical structure from the imagination of a child will forever be connected with a cancer diagnosis for one particular family. And now the challenge is on to make this molecule a reality.

Filucca Sophia Bjørnskov and Thomas Bjørnskov Poulsen

Parents who have tried bringing their kids to work have probably experienced the feeling of frantically trying to get something done while still attempting to entertain their young ones. On Wednesday 19 September 2018, one of us handed the other a molecular modelling kit together with the short — but perhaps not so simple — instruction of "please try to build something". Just a couple of minutes later, the more senior author of this essay encountered a laughing child claiming to be done with what she referred to as her 'blind chicken' (Fig. 1). Although we are fairly confident in our characterization of this structure, we should point out that neither of us has actually met a real blind chicken, so we cannot know this with absolute certainty.

These photos still elicit a feeling that is difficult to explain. The trip to the lab on that day was nothing more than a short stop — to talk to a few people and to cancel meetings — on our way to the hospital. The next day, Filucca underwent a procedure (iodine scintigraphy) that revealed a large 'cold' adenoma on the left lobe of her thyroid gland. This was the decisive event that ultimately led to a cancer diagnosis approximately three weeks later. The contrast between the gravity of this situation and the light-hearted, innocent child's play with the molecular models just a few weeks earlier was utterly surreal and explains why the memories evoked by these pictures are far from simple.

Shielding small children from the many and varied concerns of adults — and the world at large — is part of the job of being a parent, but when certain big and scary alarms sound this becomes impossible. For a seven-year-old child, the world is largely black and white and, in this world of theirs, cancer is a distant disease without nuance. As the reality of the situation began to sink in for all concerned and I experienced my child being truly scared, all I could think about was wishing that I could take her place. Just over a year later, however, we are fortunate in that the clinical scenario we are looking at appears favourable. Two surgeries in rapid succession are likely to have been curative, although we cannot say that for sure just yet.

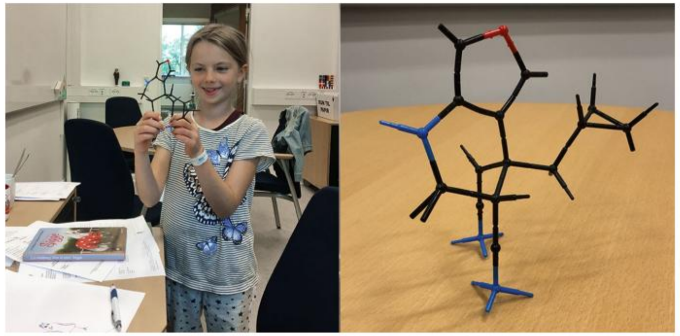


Fig. 1 | Filucca and her 'blind chicken' molecule. The product of a seven-year-old's imagination when presented with a molecular modelling kit.

You slowly learn to suppress the uncertainty associated with this disease — an uncertainty that will be with us for a long time — but it never completely goes away, of course, and it always hits you with full force right before scheduled scans and tests. One year on from where this story began, we now have first-hand experience of how having a child diagnosed with cancer irreversibly changes a family. Even following successful treatment, the road back for a child to that innocent and pure world that we associate with childhood is one that can be very difficult to find. Although our situation was certainly serious enough, watching what some of the other children in the paediatric oncology ward were going through was heartbreaking — and often numbing. The image of a small boy, tubes sticking out from under his clothes, staring with empty eyes at an untouched bowl of ice cream keeps coming back to the senior author.

Molecules are the playthings of chemists and so they can, on occasion, offer a convenient form of escapism from the macroscopic world — and the desire to inhabit a different world has been very strong at times over the past year. A more careful structural analysis of the blind chicken molecule was, therefore, perhaps inevitable. It turned out that the chicken had the amazing ability to balance on just a single atomic toe on one of its feet — and as one of us is a pretty skilled ballet dancer this insight was a source of admiration. The other leg and foot curiously turned out to be made of a material that would transform, perhaps even in a violent manner, if it ever encountered water. We have agreed that this situation is suboptimal — in fact even dangerous — for a tip-toeing blind chicken in a country where water is highly abundant. A careful dismantling of this material has therefore been performed (Fig. 2). We have also surveyed huge catalogues of known animals but it seems that the blind chicken is a rather special creature.

Although the prospects may seem bleak for families with children going through punishing treatment regimens, the chances of a positive outcome have never been better (*Cancer* **120**, 2497–2506; 2014). In any such struggle with childhood cancer, the children — and then the doctors — are heroes. Less visibly, however, modern cancer treatment rests on a platform constructed by unsung heroes who have contributed to the advanced tools needed to gather key information and to ultimately engage the disease. There are the molecular diagnostics, the contrast agents, the antibiotics and the antiemetics to list just a few examples. And then, centre stage, are the 'chemotherapeutics' or, put simply

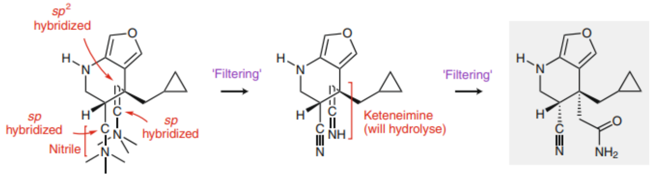


Fig. 2 | Translating the blind chicken molecule into a plausible synthetic target. The hybridization of the three carbon-atom building blocks chosen by Filucca to construct the chicken's leg is indicated (see also Fig. 1). This shows that the 'front' leg is actually a nitrile group and the 'back' leg a keteneimine. The latter is a fairly reactive group that can be expected to readily undergo hydrolysis to afford the corresponding primary amide. The 'filtered' chemical structure of the blind chicken molecule is shown in the grey box.

in the language of chemistry: molecules. The majority of these molecules have been conceived of and produced by synthetic chemists over decades and their enormous impact on the treatment statistics for childhood cancer are clear for all to see.

Reading this as a chemist, you may well appreciate what it has taken before these awesome molecules are deployed on the frontline in hospitals. From the (occasionally crazy) ideas in academia to the finely tuned research labs of the pharmaceutical industry, teams of highly dedicated scientists (including many chemists) will have invented and subsequently perfected these molecular structures to successfully overcome all of the hurdles along the way through to pre-clinical and clinical tests. And without the incredible power of synthetic chemistry to create new molecules — in our view one of the true marvels of human ingenuity — modern medicine would look very different to how it does today.

Unfortunately, synthetic chemistry is also a field of science that many people find difficult to understand. It is a fairly weird hybrid of knowing how molecules behave, puzzle solving, craftsmanship and — above all — creativity and intuition; it is a truly spectacular skill to possess at any level. There is no doubt that many of the future treatment advances in childhood cancer, as well as in medicine as a whole, will be dependent on chemists being able to think way outside of the box. As such, it is hard to come up with a better source of inspiration than the boundless imagination of a child.

Our encounter with childhood cancer came straight out of the blue, as did the blind chicken molecule. The two are now forever linked and it is impossible to forget the small molecular model still standing silent and motionless in the office. To our knowledge, no dissertations have been written about blind chicken molecules, which makes it difficult to know what to expect from them. In fact, on its first birthday, our blind chicken suddenly revealed its precocious nature when it spoke to us and told us to stop with all the adult talk and simply get down to business. This puzzling instruction was as short as the one that started this story, but we immediately understood the intention: to all synthetic chemists around the world, you are hereby challenged to make the blind chicken molecule (grey box in Fig. 2) a reality and encouraged to share your progress on social media (#blindchickenmolecule) and, ultimately, through preprints and publications.

This is a challenge that may lead to swift demonstration(s) of the advanced state of our science or it may expose methodological gaps or even prompt opportunities to think outside of the box. Maybe the blind chicken molecule can never be anything but the product of a child's imagination. Perhaps its existence will only be fleeting or — just perhaps — it could have an impact on the real world? We need to know. We like to think of this as a chemists' march for childhood cancer, which we hope can serve to increase awareness of all the important efforts made globally to further improve the lives of children with cancer.

As the chicken would say, 'The game is on.' ❐

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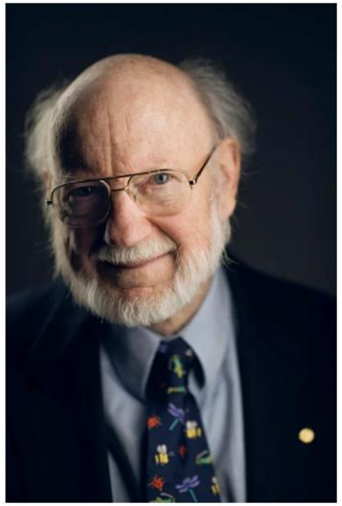
Author contributions
F.S.B. conceived of and built the blind chicken molecule without having detailed knowledge of chemistry, T.B.P. — unwittingly — caused the situation that resulted in the construction of the blind chicken molecule, analysed the structure, and wrote the article with consent and input from F.S.B.

The Nobel Prize in Physiology or Medicine 2015



William C. Campbell

Prize share: 1/4

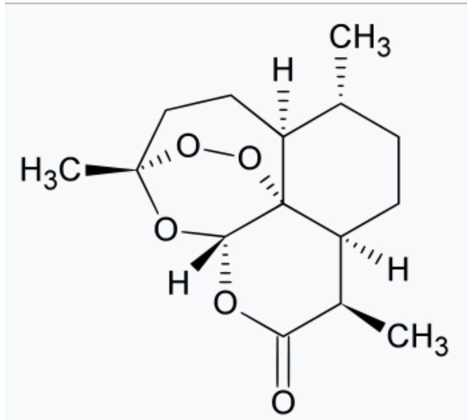
Satoshi Ōmura

Prize share: 1/4

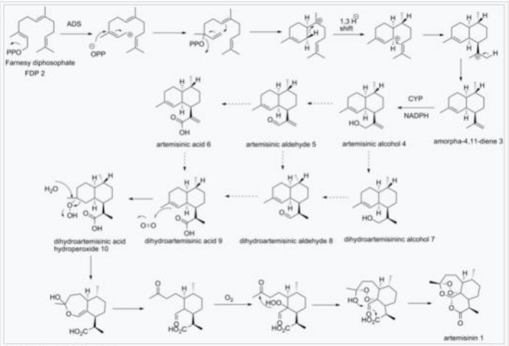
Tu Youyou

Prize share: 1/2

青蒿素

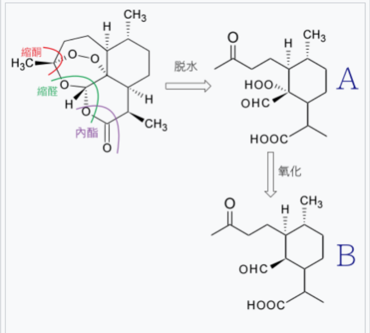


青蒿素在黄花蒿中的生物合成



青蒿素的生物合成

全合成



青蒿素全合成的逆合成分析

微生物工程菌全合成青蒿酸

The Nobel Prize in Physiology or Medicine 2015 was divided, one half jointly to William C. Campbell and Satoshi Ōmura "for their discoveries concerning a novel therapy against infections caused by roundworm parasites" and the other half to <u>Tu Youyou "for her discoveries concerning a novel therapy against Malaria"</u>

https://www.nobelprize.org/prizes/medicine/2015/summary/

What does chemistry study?

- Create entirely new substances designed to have specific properties
- Properties of substances
- The reactions that transform substances into other substances
- Understanding how and why specific chemical reactions occur
- Tailor the properties of existing substances to meet particular needs

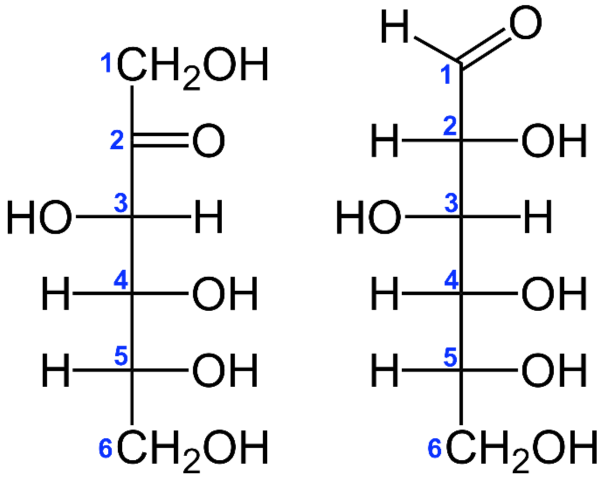
Francesco Illy founded **illycaffè** in **1933**. His **1935** invention of the **illetta**, considered the blueprint for modern espresso machines, revolutionized coffee preparation. His innovative method of packaging, based on **pressurization**, enabled illy's initial exports to **Sweden** and **Holland** during the **1940s**. Francesco Illy's method remains the standard for preserving and enhancing coffee's freshness during transport and storage.

Ernesto, Francesco's son, earned a doctorate in chemistry and joined the company in the **late 1940s**. His passion for learning gave rise to illy's formal scientific and technological research efforts, starting with an-house laboratory dedicated to coffee chemistry. In the **1950s**, he spear-headed the **company expansion into homes**, selling smaller cans of ground coffee for the first time. In **1965** he moved the company to its current **Via Flavia headquarters**, still in **Trieste**.

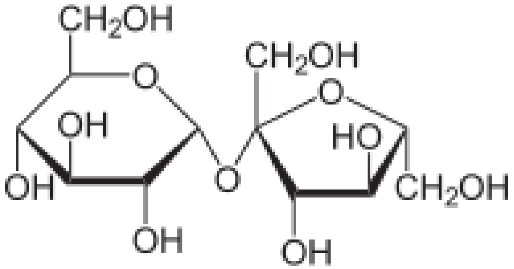
In **1974**, Dr. Illy furthered the company's lead in **coffee innovation** with **ESE**, the **first pre-measured espresso pods**, making café-quality espresso simple and easy at home or the bar. In **1988**, he introduced and patented a **photo-chromatic means to identify** the highest quality beans, one by one, right at the source.

https://www.bilibili.com/video/BV15s411R7iz

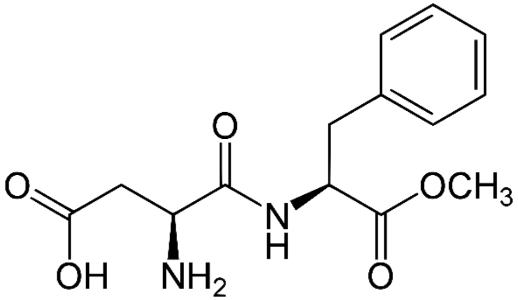
HIGH FRUCTOSE CORN SYRUP



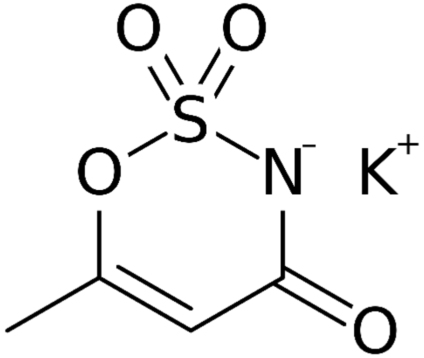
CANE SUGAR



ASPARTAME



ASPARTAME
ACESULFAME POTASSIUM



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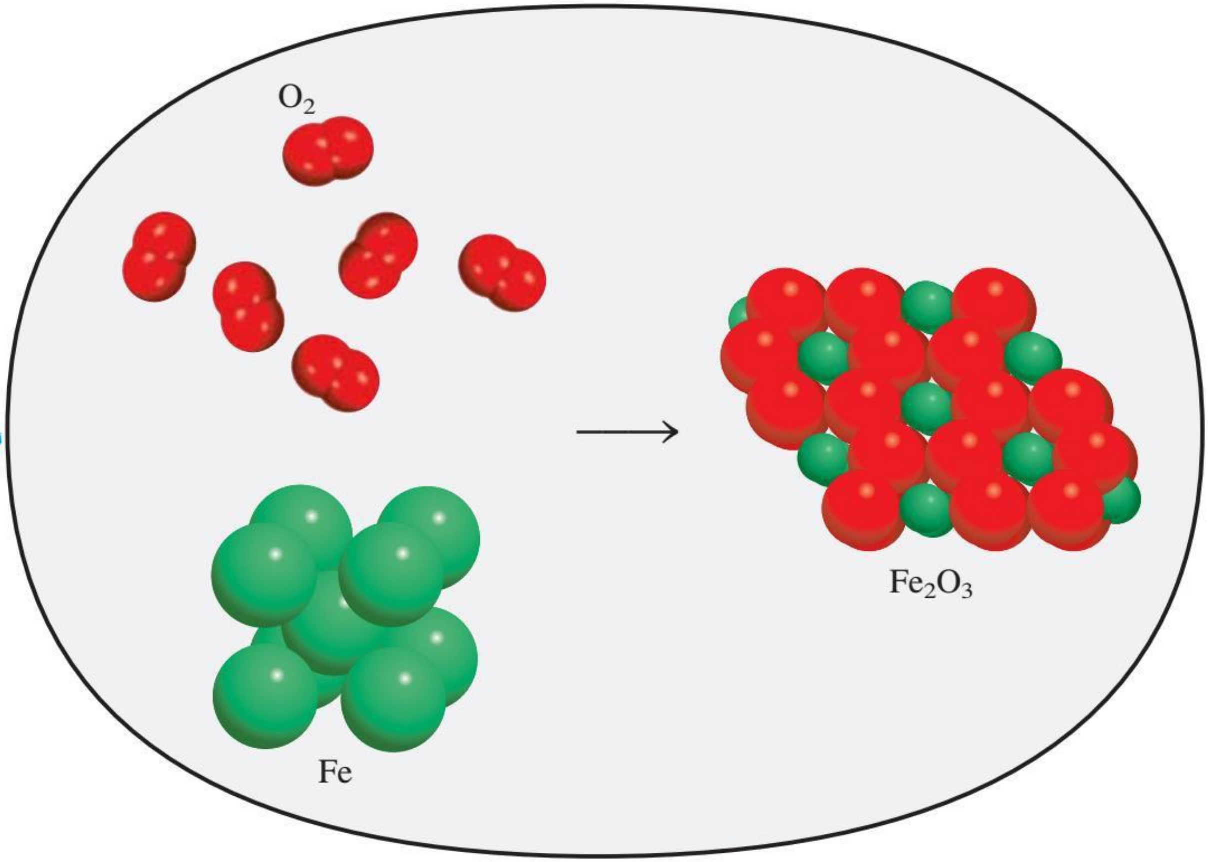
碳酸氢钠
（小苏打）
Sodium Bicarbonate
规格：500g/袋

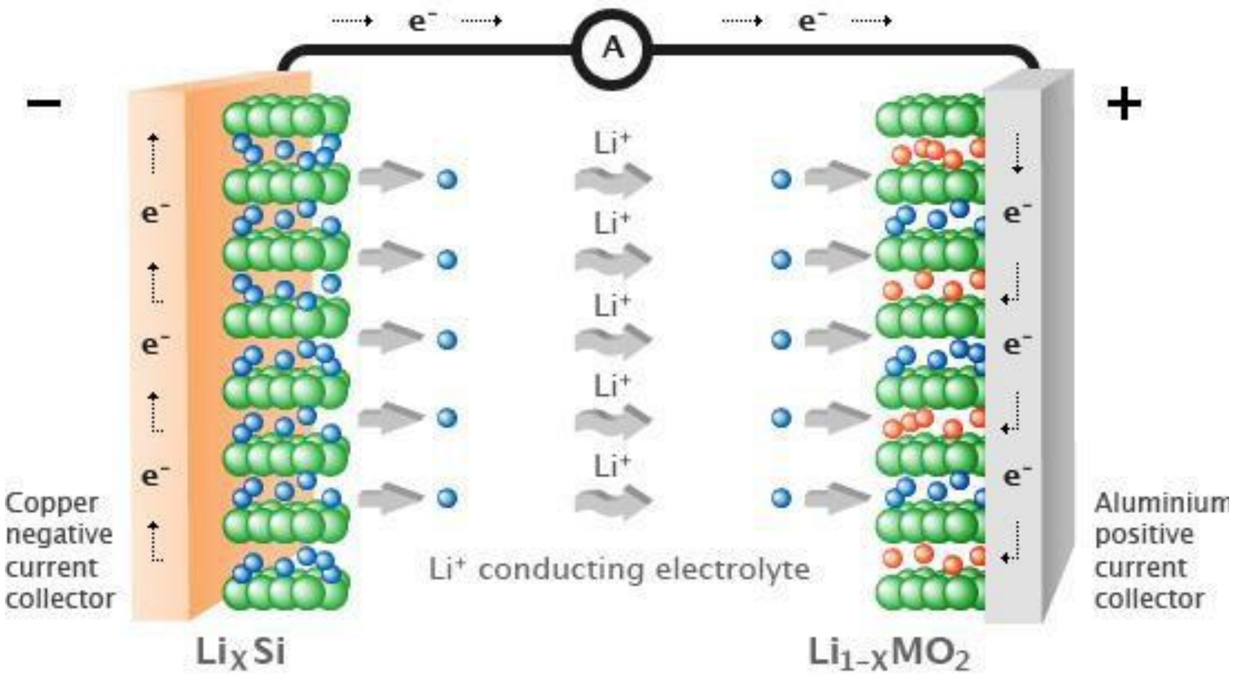
Reactions that transform substances into other substances



Reaction on positive, $LiCoO_2 = Li_{(1-x)}CoO_2+xLi^++xe^-$

Reaction on negative, $C_6+xLi^++xe^- = LixC_6$

Total reaction: $LiCoO_2+C_6 = Li_{(1-x)}CoO_2+Li_xC_6$



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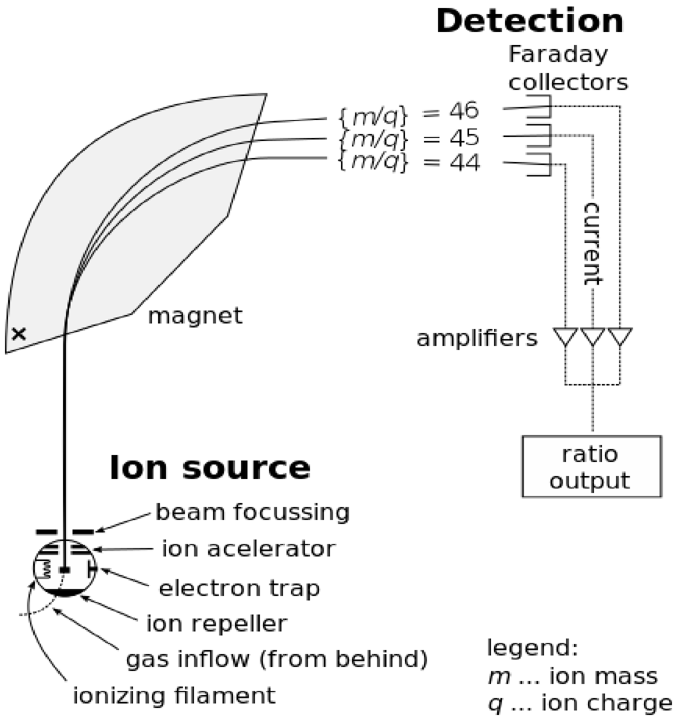
Analyze chemicals

- Quantitative analysis
- since 1900s

- Nuclear magnetic resonance (NMR)
- since 1945



- Mass spectroscopy
- since 1918

What does chemistry study?

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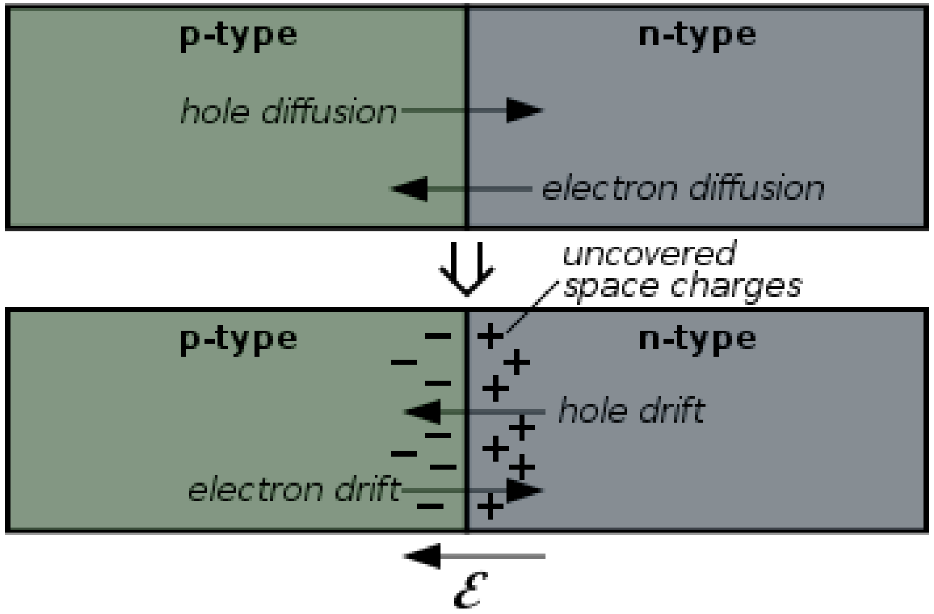
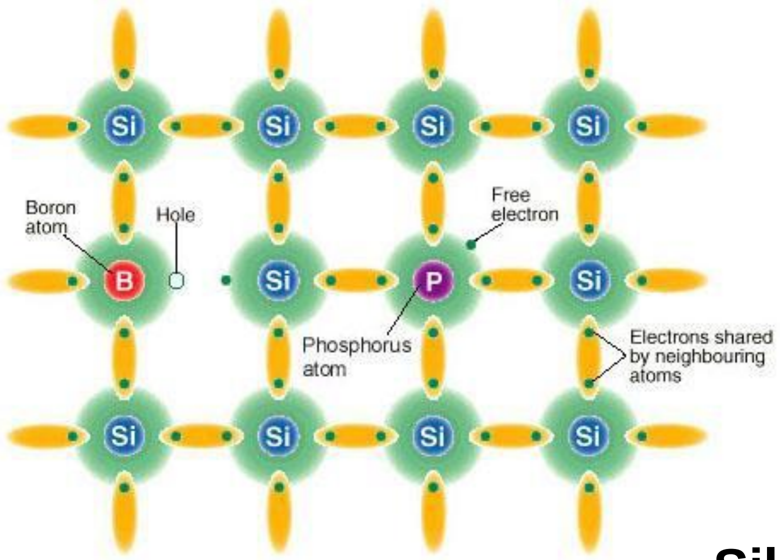
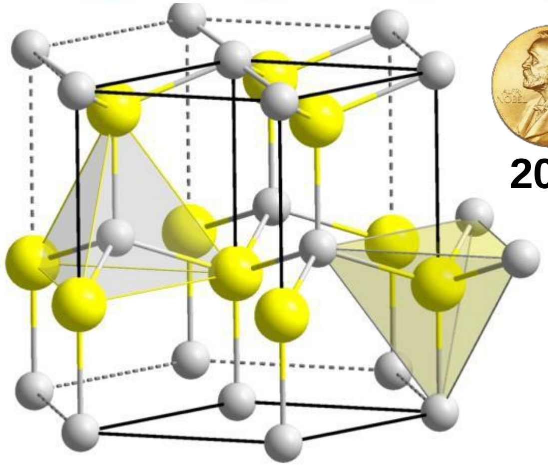
Tailor the properties of existing substances

p-n junction





Silicon



2014

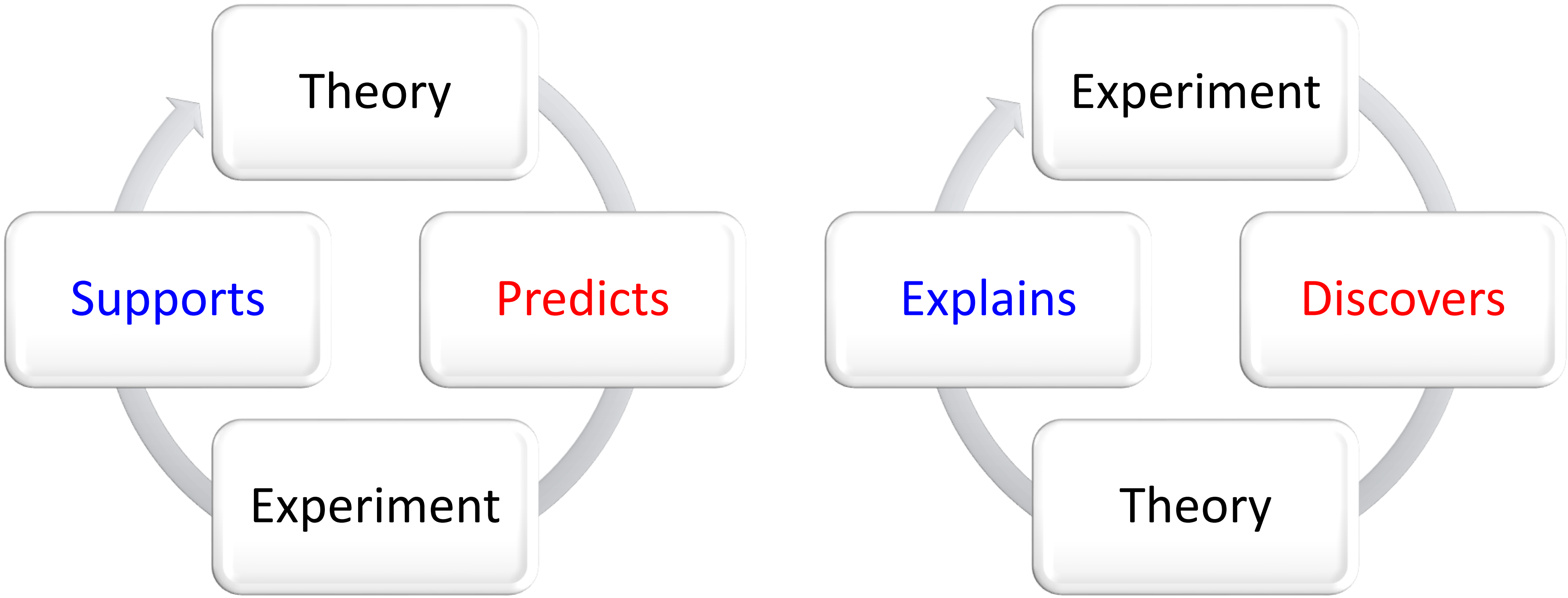
Gallium nitride
GaN

Outline

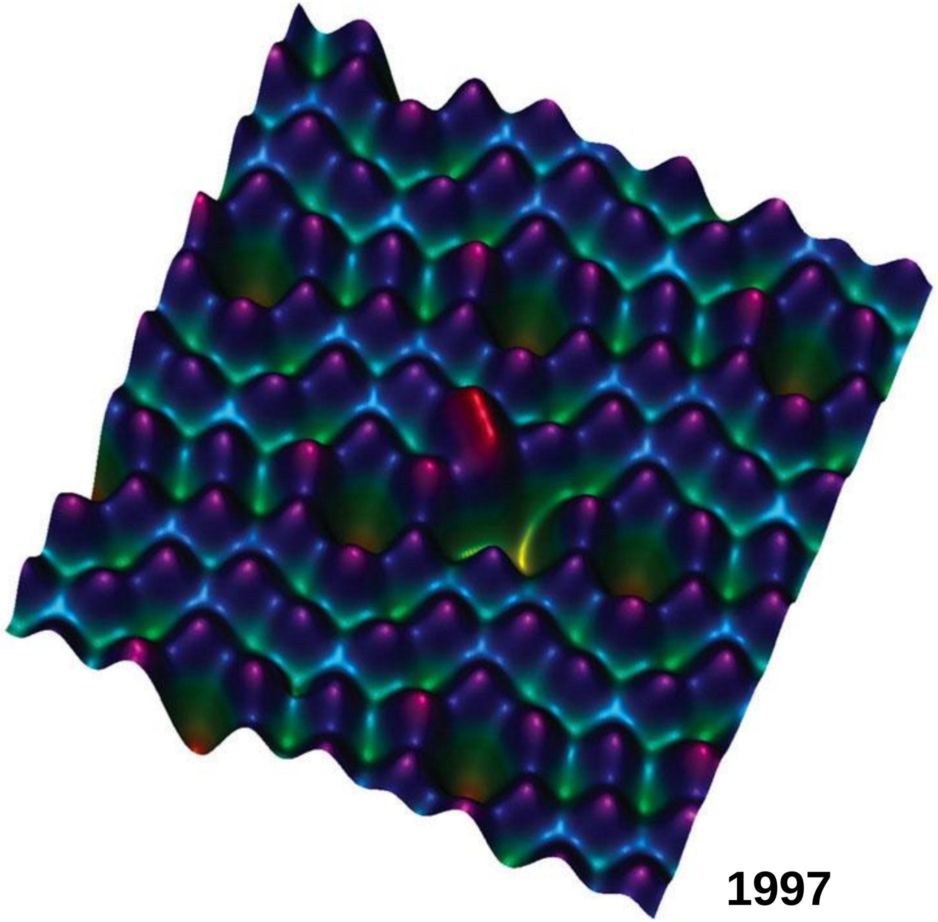
- Why is chemistry central to our life?
- What does chemistry study?
- What has chemistry achieved and what it hasn’t?
- What do we benefit from learning chemistry?

Theory vs. Experiment

Chemistry is *<u>the</u>* discipline where theory and experiment meet in harmony.

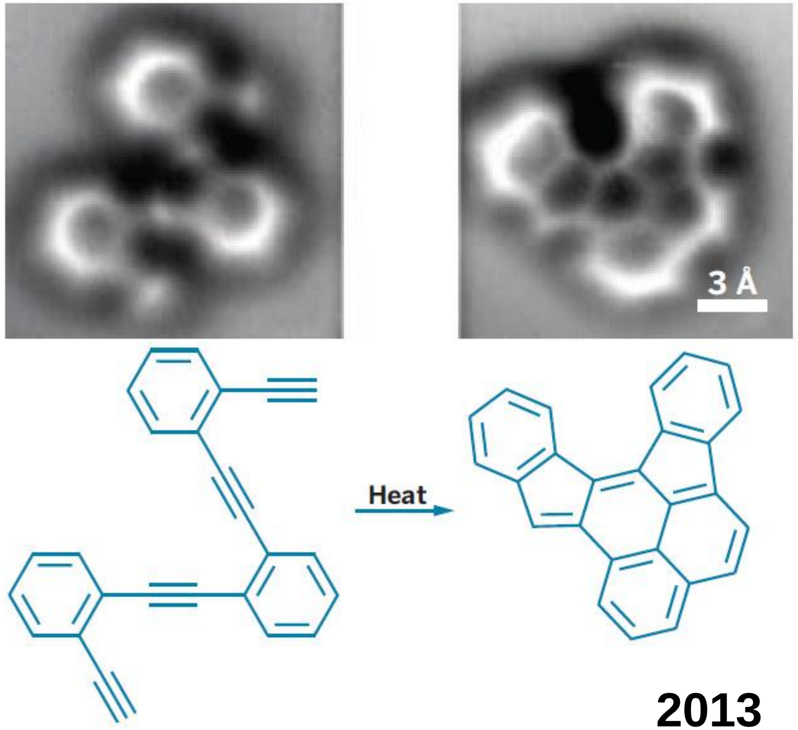


… And Focuses on Atoms and Bonds



1997

Individual atoms on Si surface



2013

The making of bonds

Outline

- **Why is chemistry central to our life?**
- **What does chemistry study?**
- **What has chemistry achieved and what hasn't?**
- **What do we benefit from learning chemistry?**

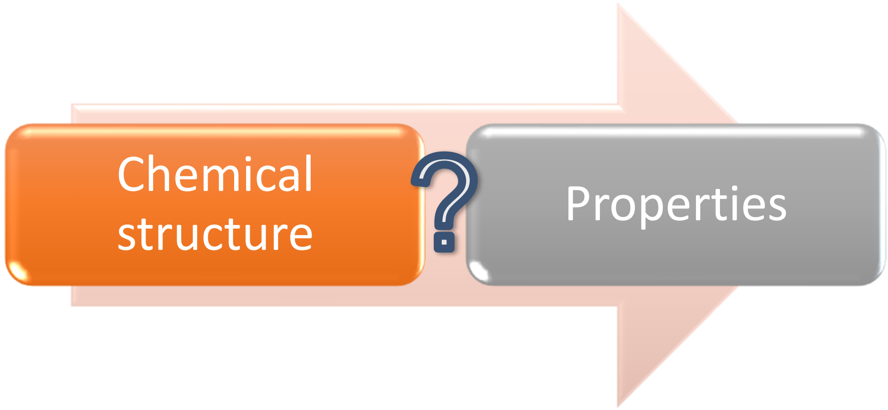
Chemistry and Mathematics

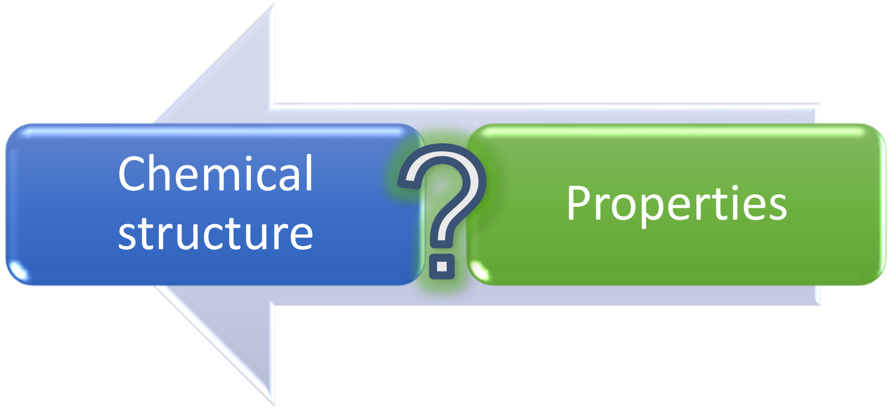
Paul A. M. Dirac
(1902–1984)

The fundamental laws necessary for the mathematical treatment of a large part of physics and the whole of chemistry are thus completely known, and the difficulty lies only in the fact that application of these laws leads to equations that are too complex to be solved.

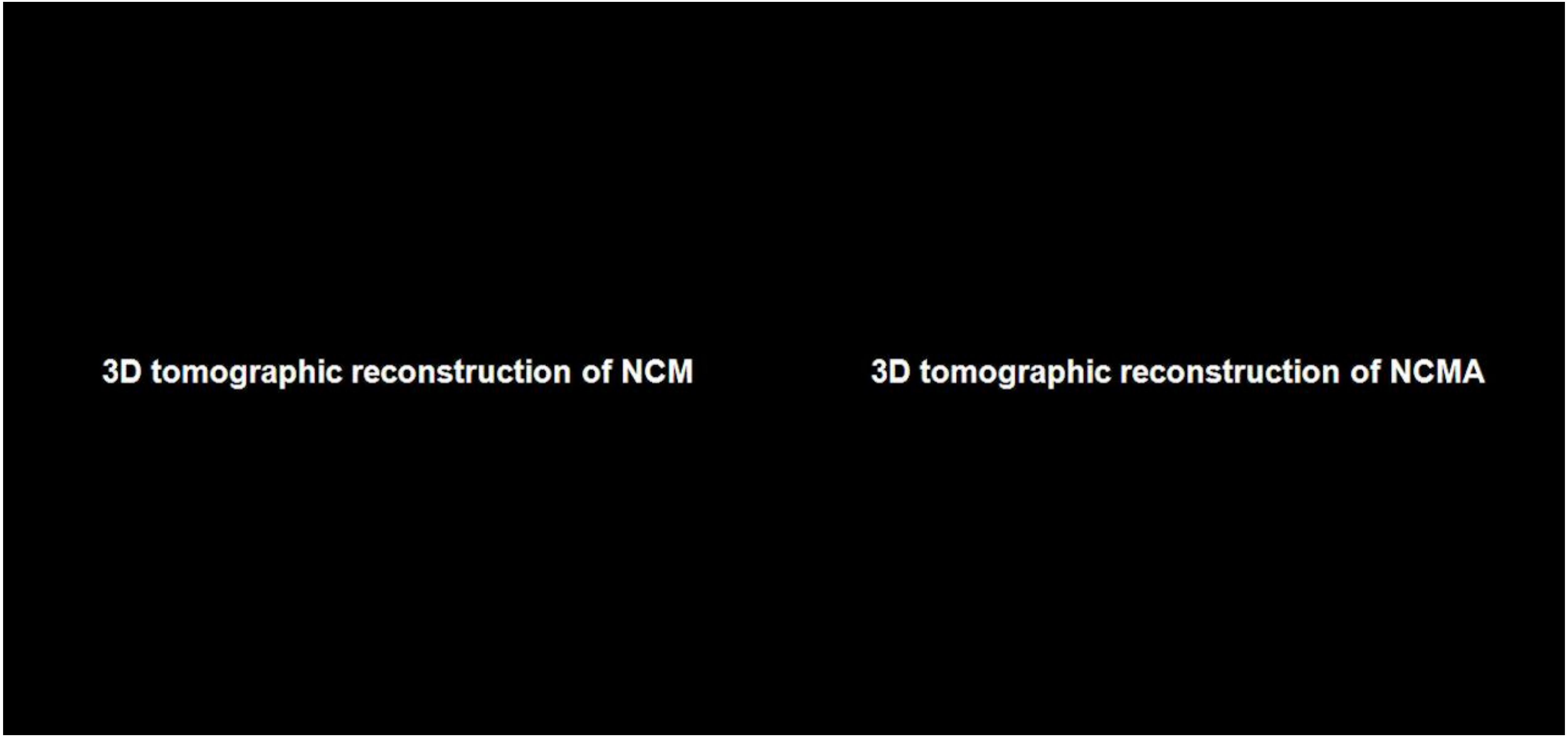
The many-electron Schrödinger Equation can only be solved approximately.

Chemistry and Materials Science

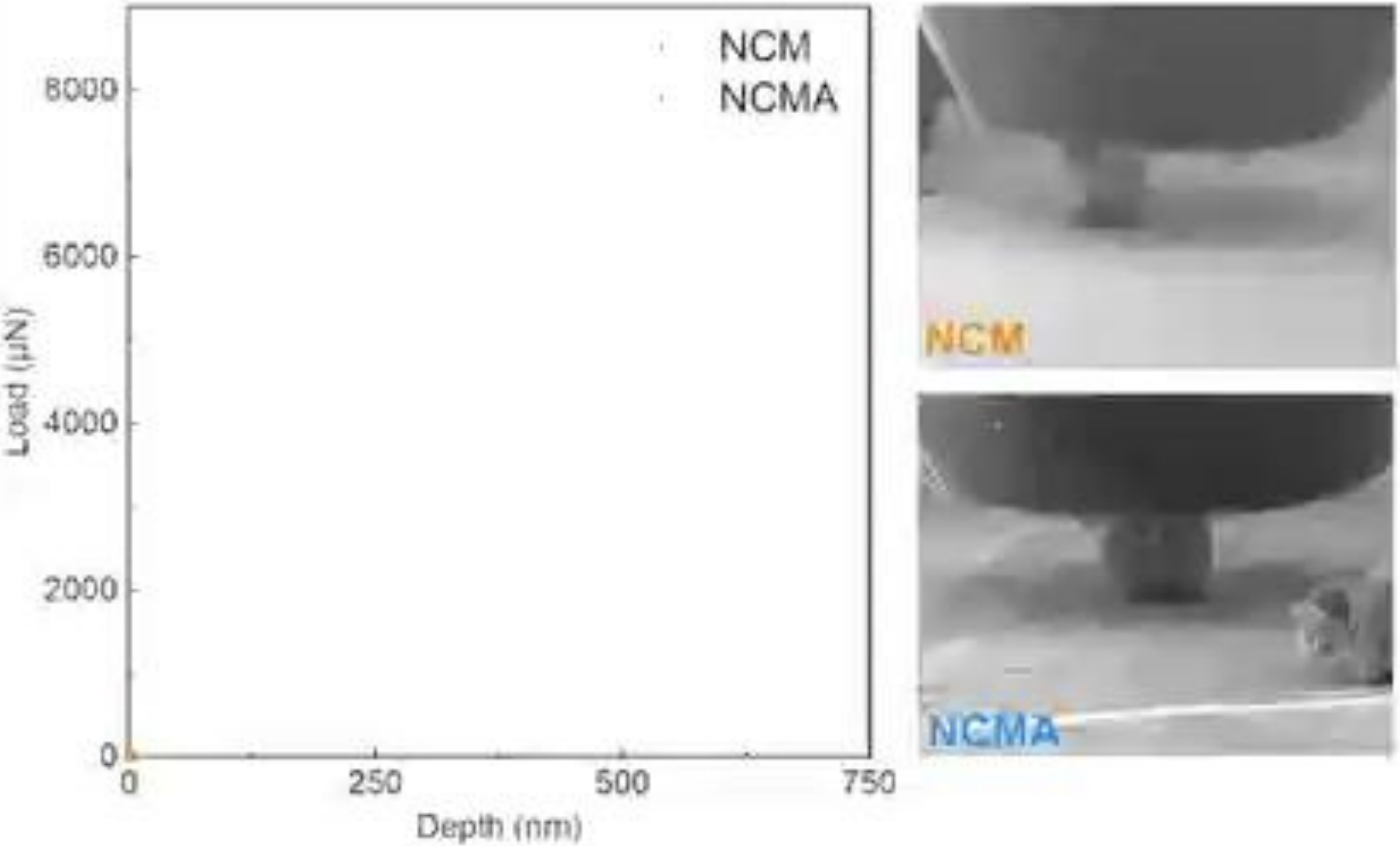




Materials Chemistry



In situ SEM mechanical characterization



Outline

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Industry

intel

Pfizer

中国石化 SINOPEC

中国石油

HUAWEI

GILEAD
Advancing Therapeutics.
Improving Lives.

PRAXAIR

TESLA

gsk GlaxoSmithKline

DOW

Boehringer
Ingelheim

APPLIED MATERIALS
FOUNDATION

Apple Computer, Inc.

NOVARTIS

DOW CORNING

Roche

DUPONT

Celanese

Lam
RESEARCH

Academia

Massachusetts Institute of Technology

SIOM

SICCAS Since 1928

SIOC 1950

SITP

National Center for Protein Science Shanghai

国家蛋白质科学中心 · 上海

SIMIT

SINAP

SIMM

Consulting / Media

SIPO
国家知识产权局

BCG
THE BOSTON CONSULTING GROUP

McKinsey&Company

NON SOLUS
ELSEVIER

WILEY

SPRINGER
NATURE

Chemists on Show

VS

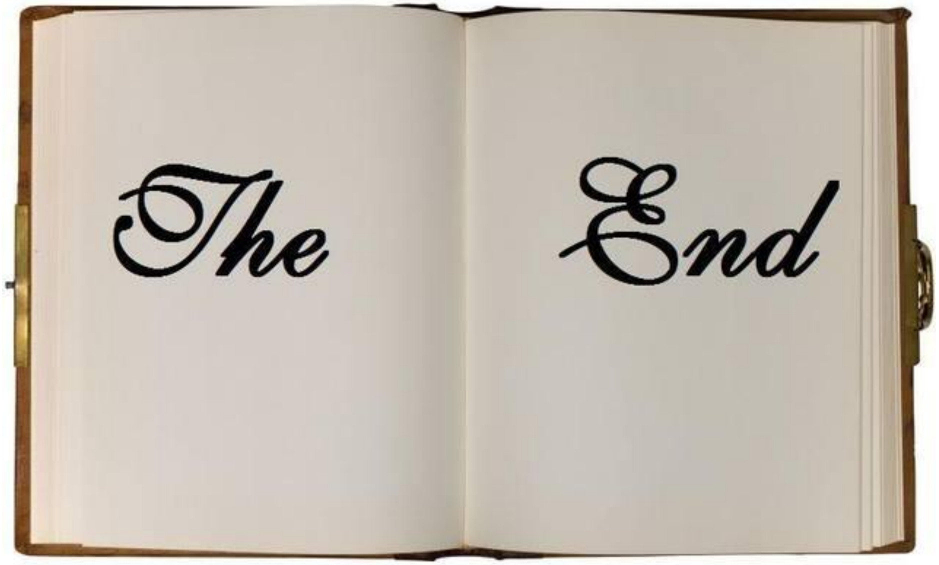
Walter White
in *Breaking Bad*

Sheldon Cooper
in *The Big Bang Theory*

Summary

- **Chemistry is about everyday life, and vice versa.**
- **Chemical theory, experiment, and applications meet in harmony.**
- **Chemistry expands its frontiers into interdisciplinary research.**
- **Chemistry enriches your life and helps you embark on a wonderful career.**

Chemistry: The Central Science



Homework: read “principle of modern chemistry”, chapter 1.